



Essay / Assignment Title: Eco-Travel Advisor

**Landa Travels, a Conversational Agent for Sustainable Tourism
Planning using the Rasa Platform**

Programme title: MSc in Artificial Intelligence

**Module: Advanced Conversational UI Design and Chatbot
Development**

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1. INTRODUCTION

Lenzen et al. (2018) estimated that tourism accounted for almost 8% of global greenhouse gas emissions. With that said, travellers have also shown that they want to travel in more responsible manner, however, there is no one platform that actually consolidates transportation, accommodation and activities and enough trustworthy information to help them make better eco friendly choices. I developed this project to address that problem, and the figure below shows how Landa Travels which is the platform I developed powered by RASA solves it.

From fragmented, unverified travel information to one transparent assistant

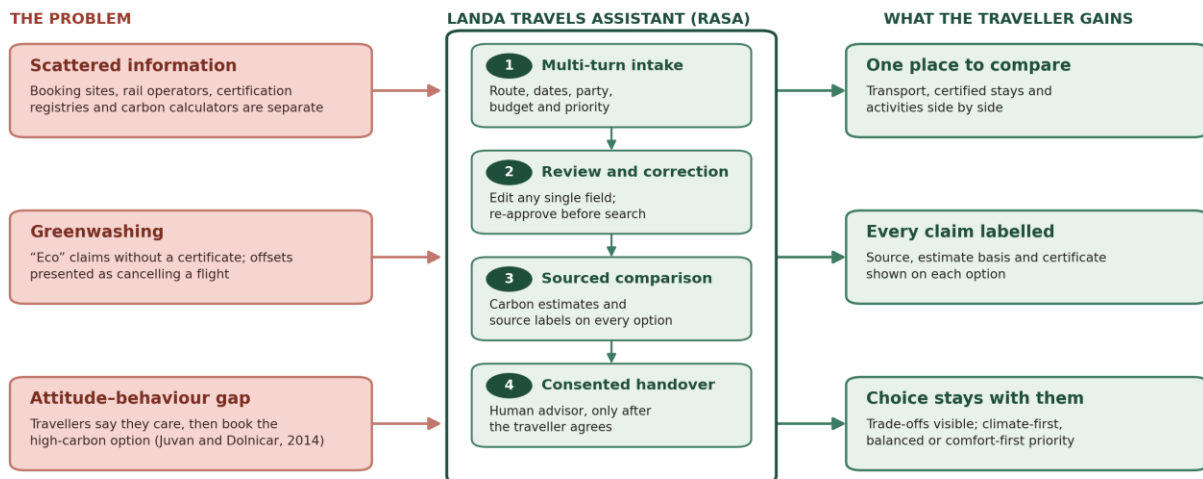


Figure 1. The problem travellers face and how Landa Travels responds

This report will summarise the research conducted, explain the functional and non-functional requirements I scoped out, present the prototype screenshots and explanations, technical implementation, the testing and deployment strategies and finally conclude with the limitations and present room for improvement.

2. RESEARCH AND TREND ANALYSIS

2.1 Review approach

For this research I looked for high-quality academic papers which were peer reviewed from 2019 and then I grouped them into themes. I also included some older foundational papers on ethics, UX and persuasive design and they were published in 2011 and 2014.

2.2 Synthesis of the evidence

Theme	Source	Method / approach	Key finding
Sustainable travel chatbots (adoption)	Pillai and Sivathanu (2020)	Survey and structural model of travellers	Usefulness, trust and perceived intelligence drive chatbot adoption
Sustainable travel chatbots (behaviour)	Majid, Tussyadiah and Kim (2024)	Qualitative service-design study	Chatbot can nudge sustainable transport after the trip
Sustainable travel chatbots (behaviour)	Majid et al. (2026)	Six-week quasi-experiment	Chatbot reminders shifted self-reported green behaviour
Sustainable travel chatbots (behaviour)	Su et al. (2026)	Five experiments incl. field study	AI agents lowered pro-environmental intent vs human staff
NLP and LLMs	Xie et al. (2024)	TravelPlanner benchmark, 1,225 tasks	GPT-4 met all constraints in only 0.6% of plans
NLP and LLMs	Bunk et al. (2020)	DIET intent and entity architecture	Lightweight, CPU-friendly joint NLU for dialogue
UX and persuasive design	Juvan and Dolnicar (2014)	Interviews with environmental activists	Strong attitude-behaviour gap on holiday
UX and persuasive design	Mertens et al. (2022); Maier et al. (2022)	Meta-analysis and bias-corrected reanalysis	Nudge effect $d = 0.43$, weak after bias correction
Ethics	Delmas and Burbano (2011)	Conceptual analysis of greenwashing	Positive claims without performance erode trust

Table 1. Synthesis of the main studies reviewed

Trust appeared a lot of times in the studies as an important factor in whether users adopt a system (Pillai and Sivathanu, 2020), whereas on the other hand greenwashing destroys that trust (Delmas and Burbano,

2011). The papers also conflict on whether automation actually helps on sustainability considering that chatbot reminders show that it improved self-reported behaviour (Majid et al., 2026) while on the other hand AI agents also lowered intentions to be more eco-friendly when comparing with human staff (Su et al., 2026). Most of the studies however are based on what people report and very few test when users are planning the trip and making decisions. Maier et al. (2022) also found that the effect of nudging became smaller after accounting for this bias. These findings guided how I thought about designing Landa Travels. I thought the platform should show users where its information is coming from and every information therefore included source and also give them information rather than push them to a choice. I also added the option to request in a human advisor when needed.

Research synthesis: from evidence to design principles

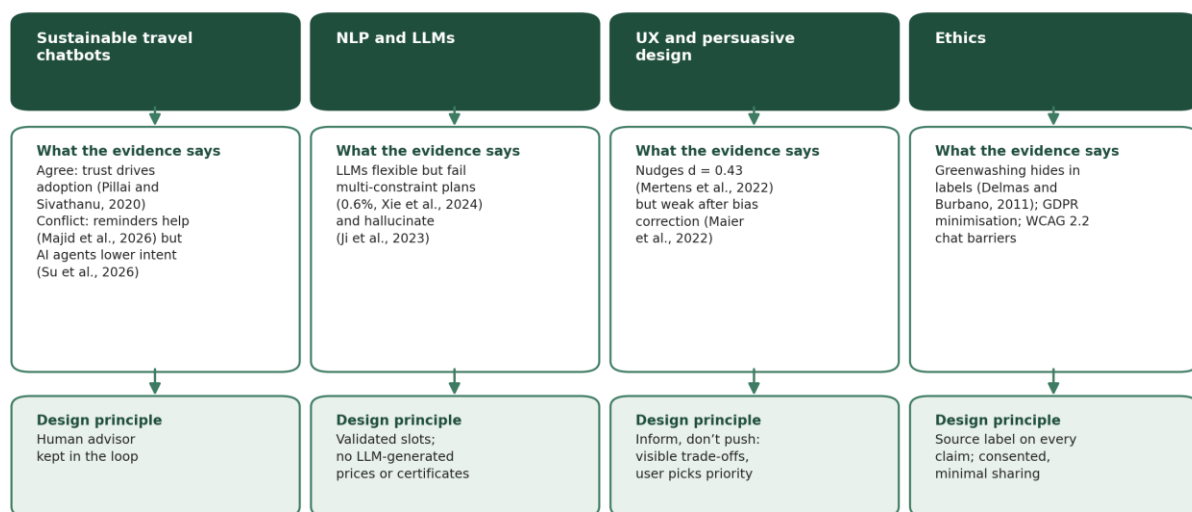


Figure 2. Research synthesis: from the four themes to design principles

2.3 Literature review and critique

Theme	Source	Key finding	Critique / limitation
Chatbot adoption	Pillai and Sivathanu (2020)	Travellers use a chatbot for planning when it is easy to use, useful, intelligent and trusted	Studies intention to adopt, not whether chatbots support sustainable choices
Sustainable behaviour	Majid, Tussyadiah and Kim (2024); Majid et al. (2026)	Chatbot reminders of past green actions changed tourists' transport behaviour	Self-reported; one destination; after the trip, not during planning when booking happens
Automation	Su et al. (2026)	Across five studies, tourists served by AI agents had lower pro-environmental intentions than those served by people	Automation may work against sustainability because users feel less morally responsible
Industry demand	Booking.com (2025)	73% of 32,000 travellers want their spending to benefit local communities	Based on stated intentions rather than actual behaviour
NLP pipelines	Bunk et al. (2020); Bocklisch et al. (2017)	DIET is small, runs on a CPU and keeps dialogue state visible through slots and forms	Less flexible with language than LLMs
Large language models	Dwivedi et al. (2024); Xie et al. (2024); Ji et al. (2023)	LLMs are flexible and expected to revolutionise trip planning	GPT-4 met all constraints in only 0.6% of plans; hallucination is a real risk for prices and certificates
Persuasive design	Mertens et al. (2022); Maier et al. (2022); Juvan and Dolnicar (2014)	Nudging closes the say-do gap, with an average effect of $d = 0.43$	Effect largely disappears after publication-bias correction; even activists excuse high-impact holidays
Interaction design	Amershi et al. (2019)	AI interactions should allow easy correction and explain recommendations	—
Ethics	Delmas and Burbano (2011); European Parliament and Council (2016); Lister et al. (2020); W3C (2024)	Greenwashing, data protection and accessibility shape responsible chatbot design	Greenwashing can happen quietly (a mapped hotel treated as certified); trip data is personal data under GDPR; chat interfaces can exclude screen-reader users

Table 2. Summary and critique of the literature reviewed

2.4 Gaps, opportunities and risks

These criticisms point to three gaps, firstly there seems to be little research on chatbots that support sustainable decisions during planning and secondly there are unreliable LLM planning and few systems actually show where each sustainability claim comes from. For those reasons my approach was supported for a slot-based Rasa with verified data over an LLM. Landa Travels also targets the third gap directly by attaching a source label and estimate basis to each and every option. The main risk is persuasive design pressuring travellers whose budget or access needs limit their choices.

3. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

I turned the assignment brief into requirements with observable acceptance criteria and recorded how each one is verified (Tables 3 and 4). Figure 3 below then shows the use cases behind the functional requirements. I added NFR6, environmental integrity as the research showed that the biggest risk for this kind of product is a claim that sounds reliable when it is actually not.

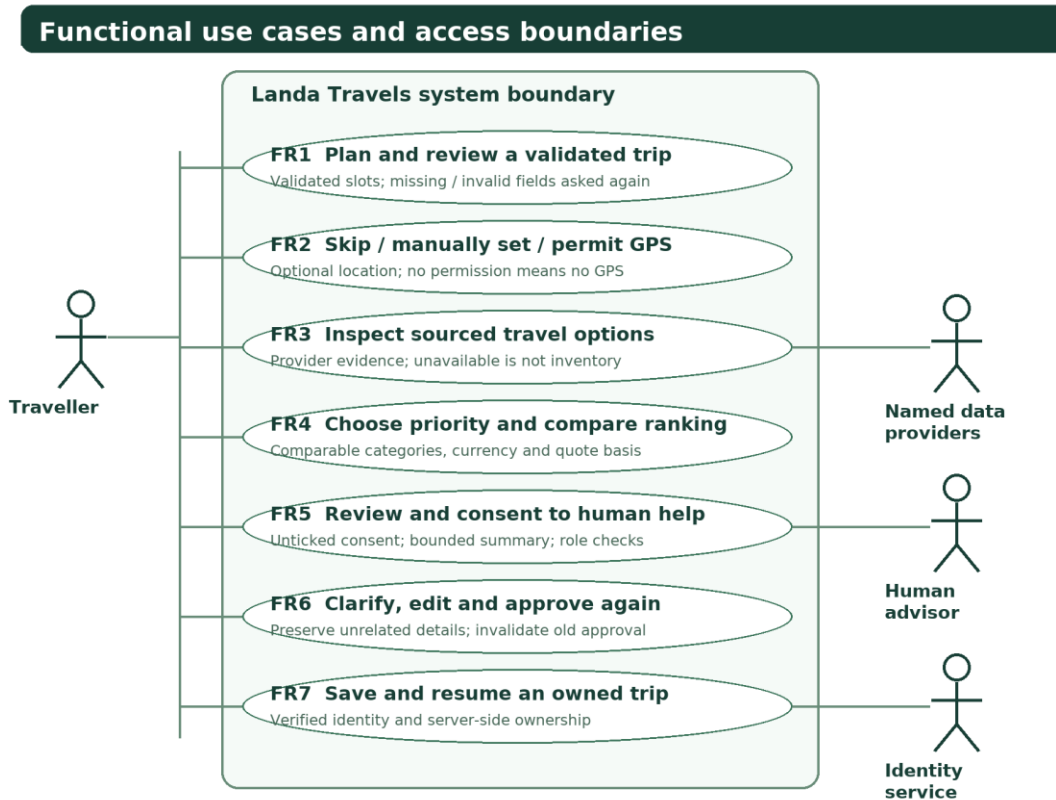


Figure 3. Use-case model: traveller, data providers, human advisor and identity service, linked to FR1–FR7

ID	Requirement	Acceptance criterion	Verification and limits
FR1	Adaptive trip intake	Form validates each field, asks only for missing or invalid values; review reflects accepted inputs	Core, source and browser tests; unseen free text can still fail
FR2	Location detection	GPS only after browser permission; manual and skip paths work without GPS	Consent branches tested
FR3	Verified retrieval	Source and estimate basis visible; outages produce an “unavailable” state	Provider tests; city coverage incomplete
FR4	Weighted ranking	Priority changes category-local weights; mismatched currencies never compared	Ranking tests and documented weights
FR5	Human escalation	Nothing sent before consent; role-protected advisor queue	Handover and ownership tests; replies asynchronous
FR6	Error recovery	Clarify the field, keep unrelated details, re-seek approval	Core tests and user testing
FR7	Save and resume	Saved trip restored; another identity cannot read it	Authorisation and hosted tests

Table 3. Functional requirements

ID	Requirement	Target and check	Status
NFR1	Usability	Plain prompts, review and correction, explanations; task-based testing	Partly met: self-testing and one additional user (6.4); larger study still to run
NFR2	Reliability	Same behaviour on desktop and mobile; provider failure never invents inventory	Partly met: fixed cases pass; no load test
NFR3	Latency	Under 3 s from send to visible reply on critical turns	Met on 29 warm turns; cold start and load unmeasured
NFR4	Accessibility	Text alongside colour, keyboard controls, announced status messages	Partly met: no full screen-reader audit
NFR5	Privacy (GDPR)	Owner-only data, consented bounded handover, 30-day conversation expiry	Controls verified; not a legal compliance audit
NFR6	Environmental integrity	Every estimate shows its basis; certificates only from an official registry; offsets separate	Met

Table 4. Non-functional requirements

4. CONVERSATIONAL PROTOTYPE AND INTERACTION DESIGN

4.1 Method

I built the product directly in React and drew the flows as diagrams, because the assignment brief asks for UI elements wired to real bot data and a Figma mockup would have then hidden the integration problems such as quick reply labels being misclassified.

4.2 Dialogue scenarios and branching

I designed three scenarios that test different parts of the conversation (Table 5, Figure 4). All those three share the same backbone of intake, review, correction and optional human help, but the questions and responses change with the traveller's needs, priority and budget.

Scenario	Traveller's situation	How the dialogue adapts	What it demonstrates
City break	Two adults, Berlin to Paris, a few days, fixed budget	Fills several details from one sentence; asks only for what is missing; handles a change of departure city	Efficient intake and correction
Rural eco-tour	Walking trip, one traveller needs step-free access, declines GPS	Asks about access needs; works from a typed location; says plainly when a mapped stop does not prove step-free access	Honest limits and inclusion
Business trip	Fixed meeting date; time, price and carbon conflict	Shows the trade-off and asks which constraint can move instead of quietly relaxing one	Transparent trade-offs

Table 5. Designed dialogue scenarios

Three designed dialogue scenarios

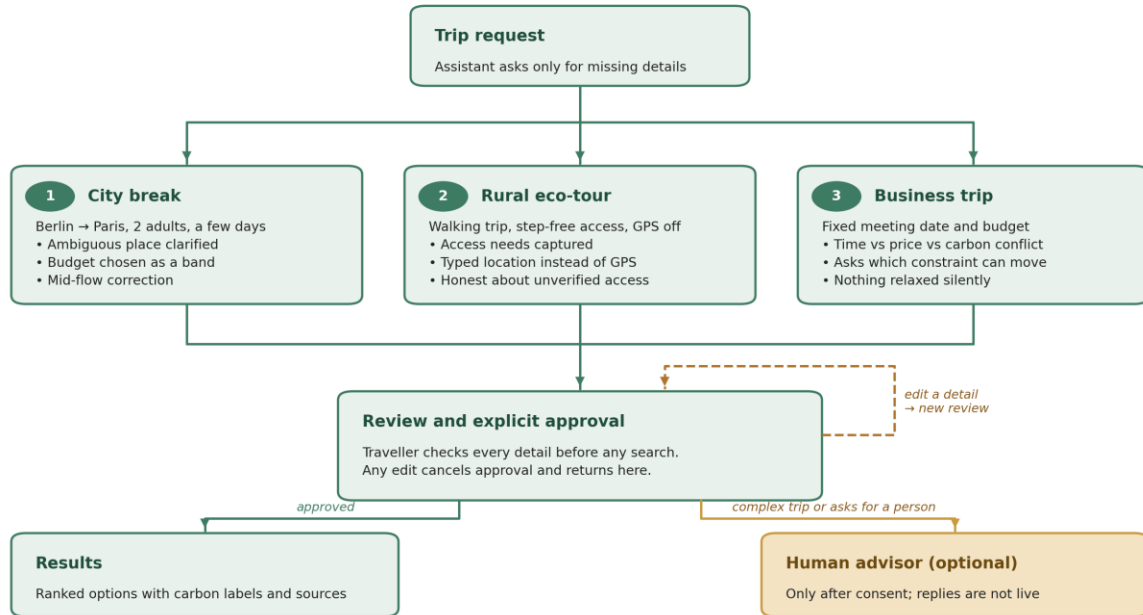


Figure 4. The three designed scenarios sharing one intake, review and handover backbone

4.3 State model and correction

Figure 5 below shows the state model behind the planner/ chatbot.

Planner state and correction behaviour

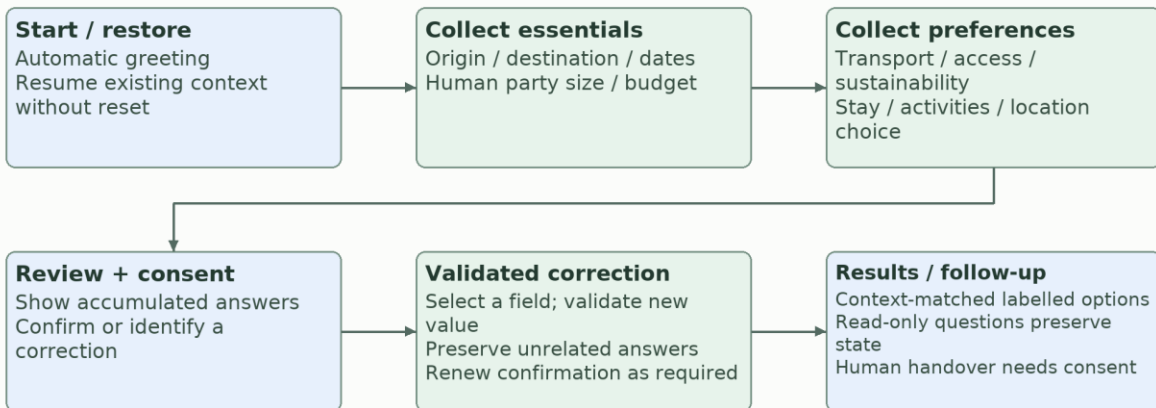


Figure 5. Planner state model with field-specific correction and re-approval

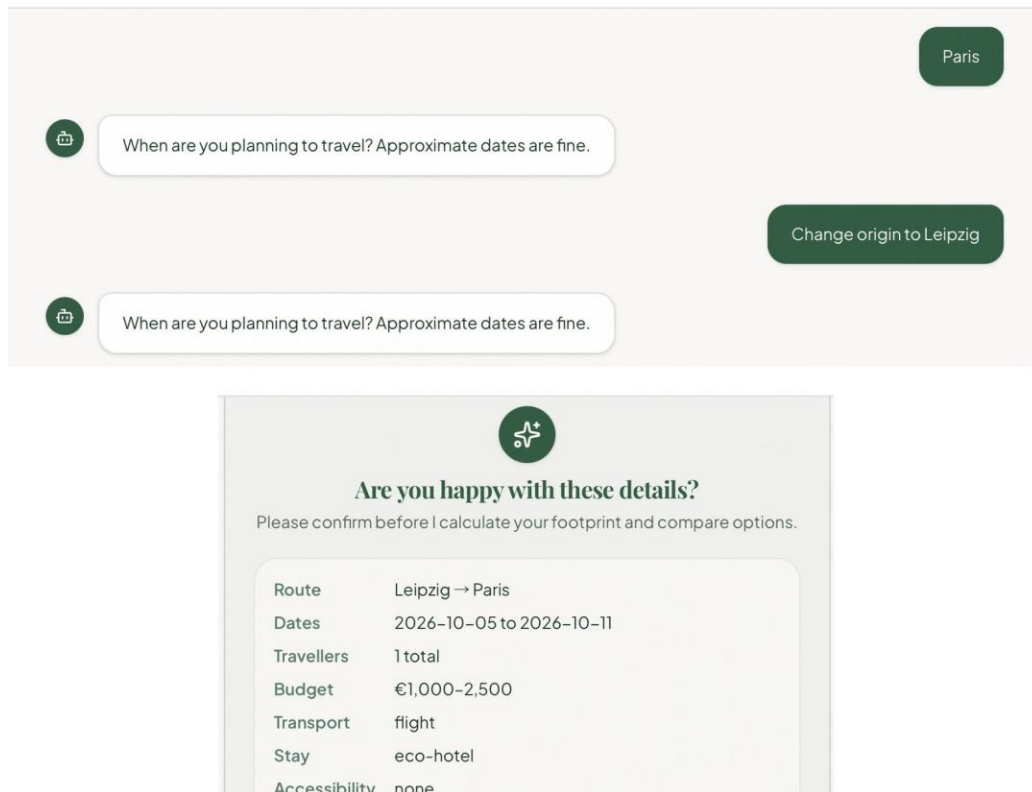


Figure 6. Changing only the departure city in the chat, then the review card that asks for explicit approval

4.4 Interface elements

The public site shown by Figure 7 lets travellers start planning without an account and then Figure 8 shows the planner. Quick replies are generated from the assistant's responses such as the choices in Figure 9 and then selecting one sends the structured value rather than the button text. Each option also carries a green, amber or red carbon label with its kilogram value. Figure 10 below shows this on a Leipzig to Paris trip and a daytime rail is labelled low impact with 13.4 kg CO₂e, the test-mode flight moderate, and a high-impact item adds a red high-carbon warning, so the traveller can see actual the trade-off. The layout reflows on mobile (Figure 11).

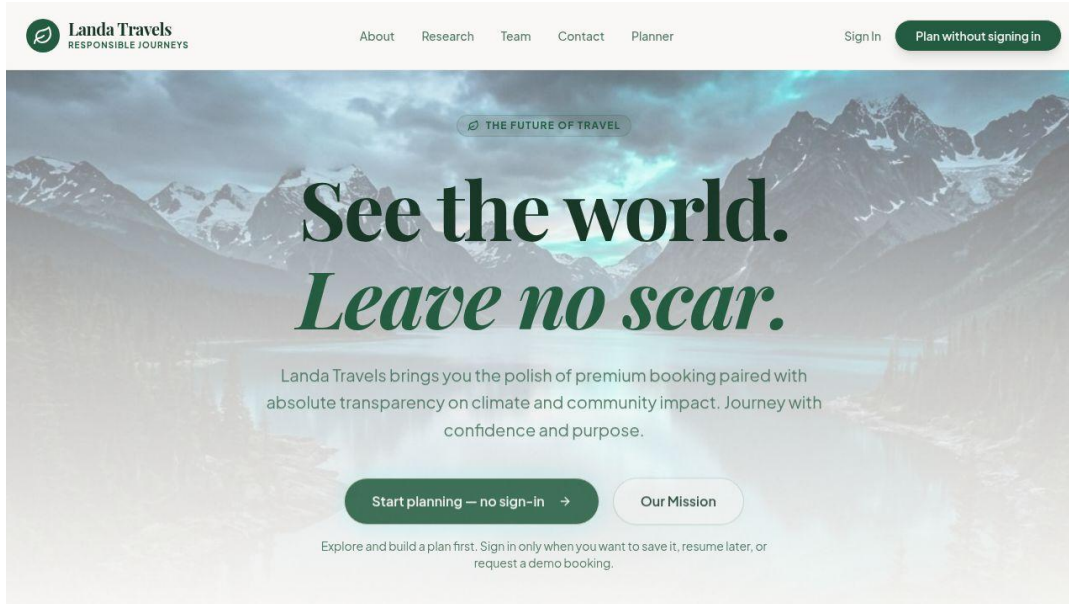


Figure 7. Public homepage with entry into anonymous planning

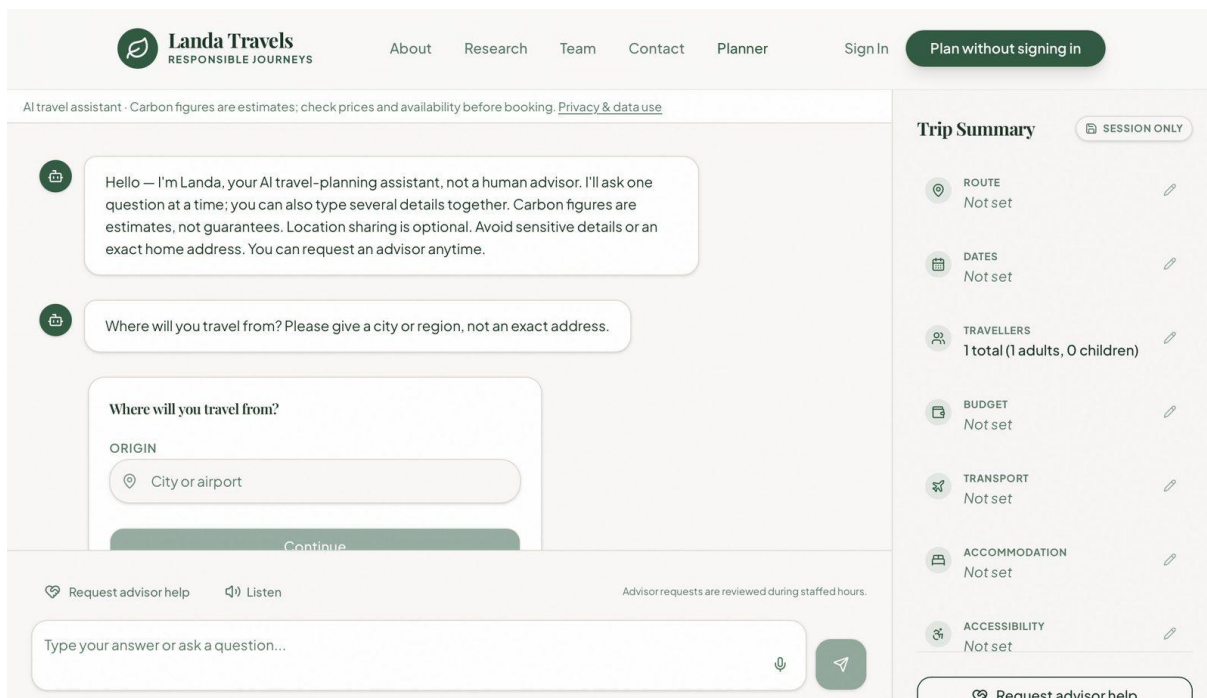


Figure 8. Planner opening with AI disclosure, the first question and the live trip summary

Choose your travel dates

Select departure and return separately.

DEPARTURE

RETURN

5. Oct 2026

11. Oct 2026

My return date is flexible

Continue

What total budget should I work within?

Under €1,000

€1,000–2,500

€2,500–5,000

€5,000+

Continue

How do you prefer to travel?

Train
Lower impact, scenic

Bus / Coach
Cost effective

Flight
For long distances

Compare all
Show practical alternatives

How important is minimizing carbon impact?

Climate-first
Prioritise lower-carbon options

Balanced
Good eco-choices within reason

Comfort-first
Prioritise convenience and comfort

Continue

Figure 9. Guided questions with quick replies for dates, budget, transport and sustainability priority

TRANSPORT

☐ Include in advisor request

low impact

Daytime rail

Paris

Approximate return journey over 1180 km. Indicative fare is per traveller; budget ranking considers the whole party. Confirm the actual route, timetable, fare, and occupancy.

Why this ranked: balanced comparison: carbon 45%, quoted price 30%, preference 10%, convenience 15%. Journey time unknown; no speed or accessibility advantage assumed. Quoted component price, not the total itinerary cost.

PRICE SHOWN

€142 per traveller · indicative

Pricing can change; confirm directly with the source.

CARBON

13.4 kg CO₂e

Estimate basis not specified

TRANSPORT

☐ Include in advisor request

moderate impact

Duffel Airways

Leipzig/Halle Airport → Paris

Approx. 190 min

Duffel TEST MODE return offer. Route-average ClimaTiq estimate is not aircraft-

OFFSET

☐ Include in advisor request

high impact

Verified climate contribution

Paris

Only after reducing the trip footprint, consider a programme that publishes its methodology, additionality evidence, registry serial numbers, and retirement records. The displayed 196.9 kg CO₂e is the footprint to address, not emissions erased by payment.

Why this ranked: balanced comparison: carbon 45%, quoted price 30%, preference 10%, convenience 15%. Journey time unknown; no speed or accessibility advantage assumed. Quoted component price, not the total itinerary cost.

High-carbon option

This option has a high estimated impact: 196.9 kg CO₂e. Consider a lower-carbon alternative if practical.

Figure 10. Recommendation cards with low, moderate and high impact labels, ranking explanation and high-carbon warning

 **Landa Travels**
RESPONSIBLE JOURNEYS

Trip Planner

VIEW SUMMARY

AI travel assistant · Carbon figures are estimates; check prices and availability before booking. [Privacy & data use](#)



Hello — I'm Landa, your AI travel-planning assistant, not a human advisor. I'll ask one question at a time; you can also type several details together. Carbon figures are estimates, not guarantees. Location sharing is optional. Avoid sensitive details or an exact home address. You can request an advisor anytime.



Where will you travel from? Please give a city or region, not an exact address.

Where will you travel from?

ORIGIN

 City or airport

 Request advisor help

 Listen

Type your answer or ask a question...





You can type or use the microphone. Voice stays in your browser until you send the transcript.

Figure 11. Planner at mobile width (402 px)

[illegible]

Advisor request submitted

Your request is saved in the advisor inbox. Check its status here; an advisor can also reply to your supplied email. Inbox email notification is not confirmed; the saved request is still available to advisors.

Figure 12. Human-help panel with what will be stored, the unticked consent box, and the confirmation after submission

5. TECHNICAL IMPLEMENTATION AND NLP INTEGRATION

5.1 Architecture

Figure 13 shows below show the full system layers. Figure 14 shows the same flow as data moving between processes and stores.

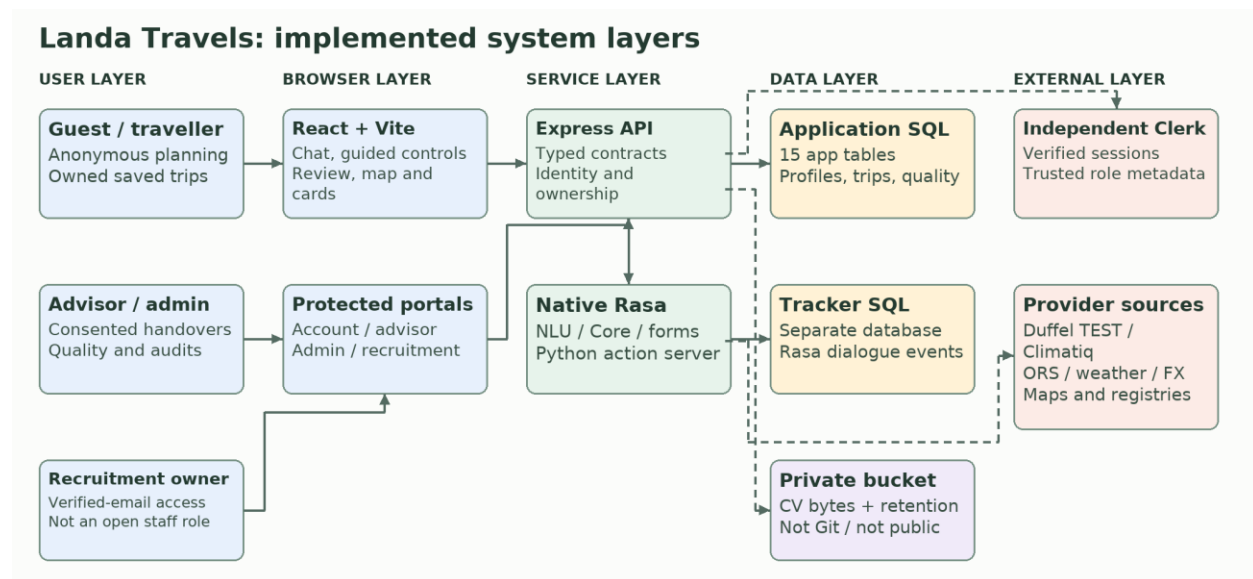


Figure 13. Implemented system layers and trust boundaries

Data-flow model: input, processing and persistence

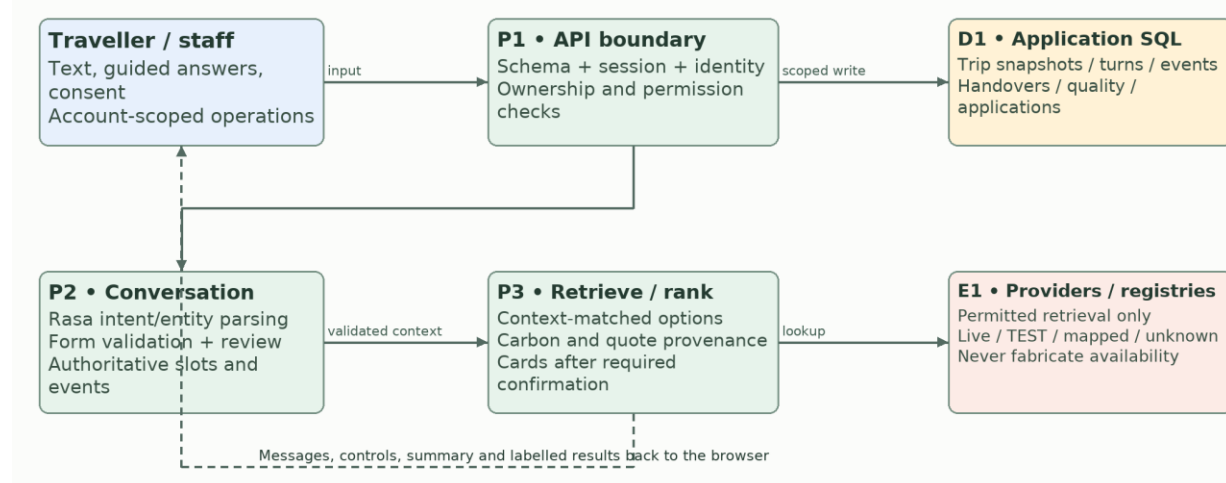


Figure 14. Data-flow model of input, validation, conversation, retrieval and persistence

5.2 NLU and dialogue management

The NLU pipeline follows the assignment brief's recommendation of DIET with spaCy features (Table 6, Figure 15). I also added character n-grams for misspelt place names and a custom TravelEntityExtractor

that them normalises values such as “two adults” into counts. I however did not use DistilBERT for this project because it was optional and also because DIET already met the latency target on a CPU.

Component	Key settings	Reason
SpacyNLP, tokenizer, featurizer	en_core_web_md, mean pooling	Dense word vectors on CPU
Lexical, Regex, CountVectors	Word counts plus char_wb 3-5 grams	Handles typos and rare place names
DIETClassifier	100 epochs, entity recognition on	Joint intent and entity model
FallbackClassifier	Threshold 0.35, ambiguity 0.05	Catches low-confidence turns
TravelEntityExtractor (custom)	Runs last	Normalises counts, dates and places
MemoizationPolicy / RulePolicy	History 5 / Core fallback 0.4	Exact paths and fixed rules
TEDPolicy	History 8, 80 epochs	Generalises across story variations

Table 6. Rasa configuration (Rasa 3.6.21, Python 3.10)

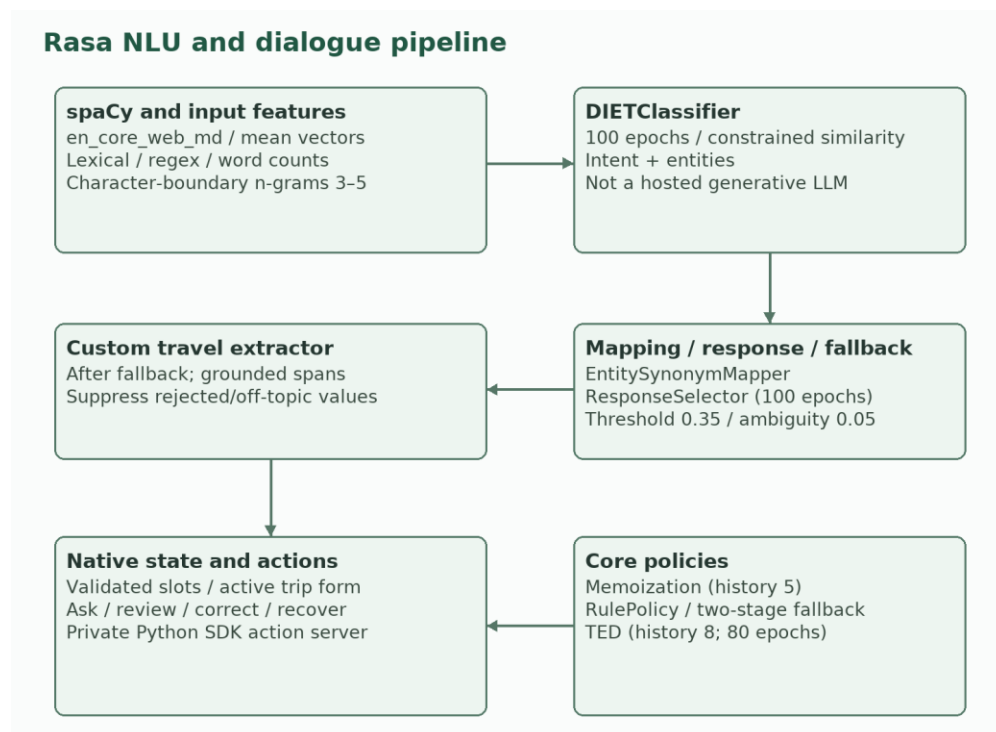


Figure 15. NLU and dialogue pipeline, with the custom extractor after fallback

Slots persist the trip details across turns and the validators reject impossible values such as past dates. Fallback handling differs slightly from the brief's wording. Outside the trip form, low confidence turns go to `action_two_stage_fallback`, which re-prompts with quick reply options and then offers human help. Inside the form I then used a form-specific clarification instead. The reason for this was technical as Rasa runs only one active loop at a time and the two-stage fallback is itself a loop so triggering it mid-form deactivated the trip form and lost the collected answers

Figure 16 below shows how NLU output, policies and validation combine into slot events.

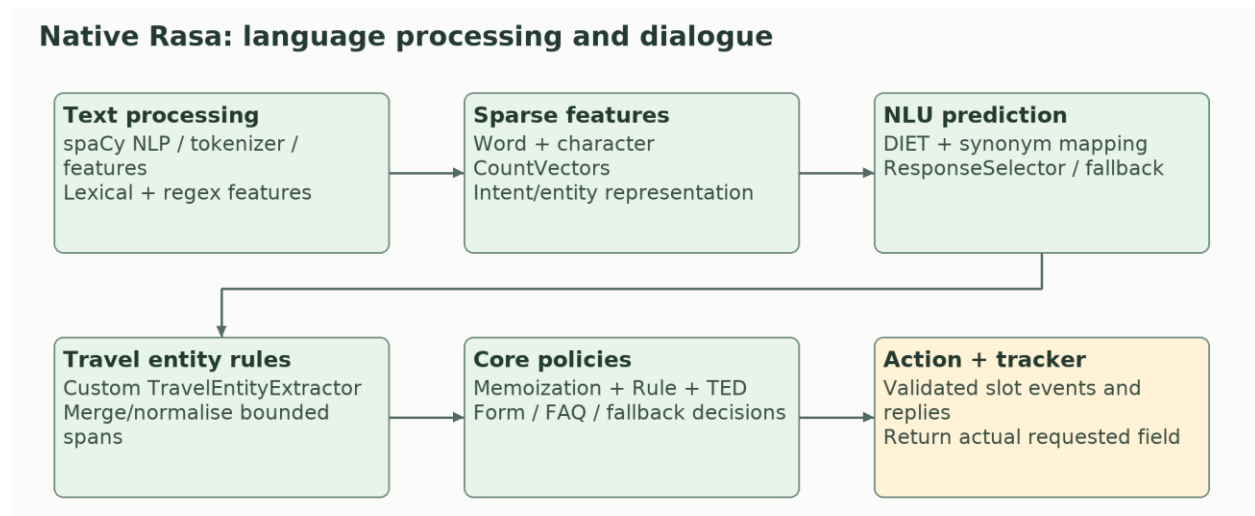


Figure 16. Rasa NLU processing, Core policy decisions and validated action events

5.3 Custom actions and external APIs

The carbon estimates come from the Climatiq Basic Estimate endpoint per transport mode and distance, and the flights come from Duffel in test mode. Amadeus was replaced because its self service APIs closed on 17 July 2026 (Crotty, 2026) and thus that was not used as per the assignment brief. Hotel certification comes only from the EU Ecolabel register and is never from matching a map name. The 30 September 2026 snapshot held 57 certified hotels in Paris, three in Berlin and one in Lisbon. Every provider call has a timeout and returns an explicit unavailable state on failure (Listing 1). Figure 17 traces one turn through these services, and Figure 18 shows how provider data becomes a labelled option.

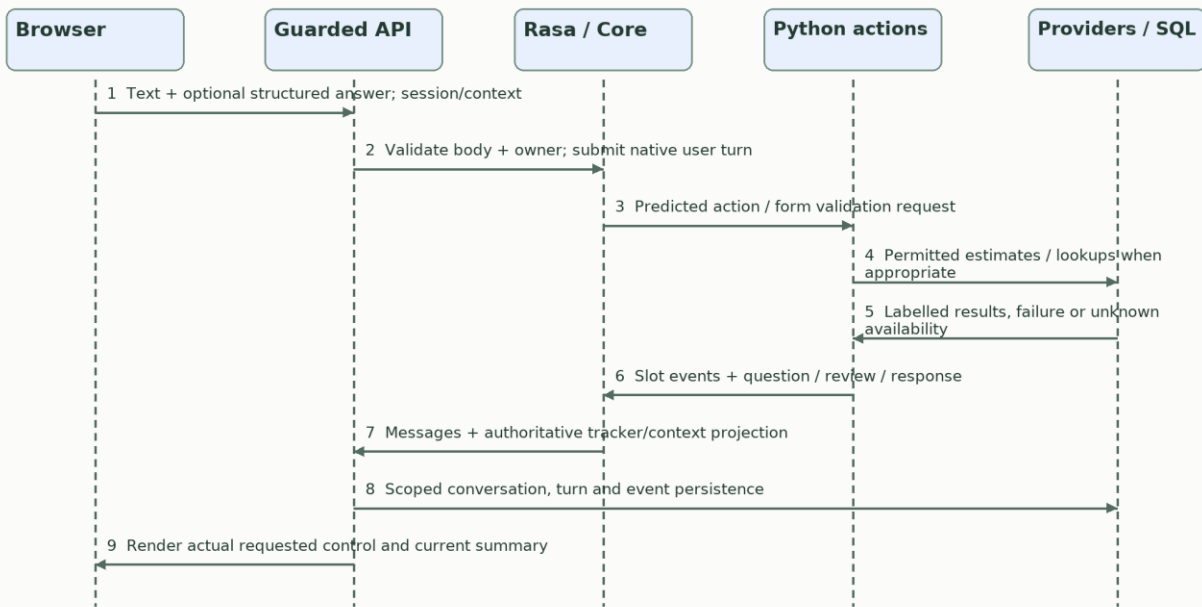
```

435 def _request_json(
436     method: str,
437     url: str,
438     *,
439     headers: dict[str, str] | None = None,
440     data: dict[str, Any] | None = None,
441     timeout: float = 2.3,
442 ) -> dict[str, Any] | None:
443     try:
444         response = requests.request(
445             method, url, headers=headers, json=data, timeout=timeout
446         )
447         response.raise_for_status()
448         return response.json()
449     except (requests.RequestException, ValueError) as exc:
450         logger.warning("External API request failed: %s", exc)
451         return None

```

Listing 1. API error handling in `rasa-bot/actions/actions.py` (lines 435–451)

Sequence diagram: one assistant turn



A guided answer is submitted on a real user turn and validated by Rasa. Network success is not proof of a correct intent, state

Figure 17. Sequence of one assistant turn across browser, API, Rasa, actions and providers

Evidence and provider provenance

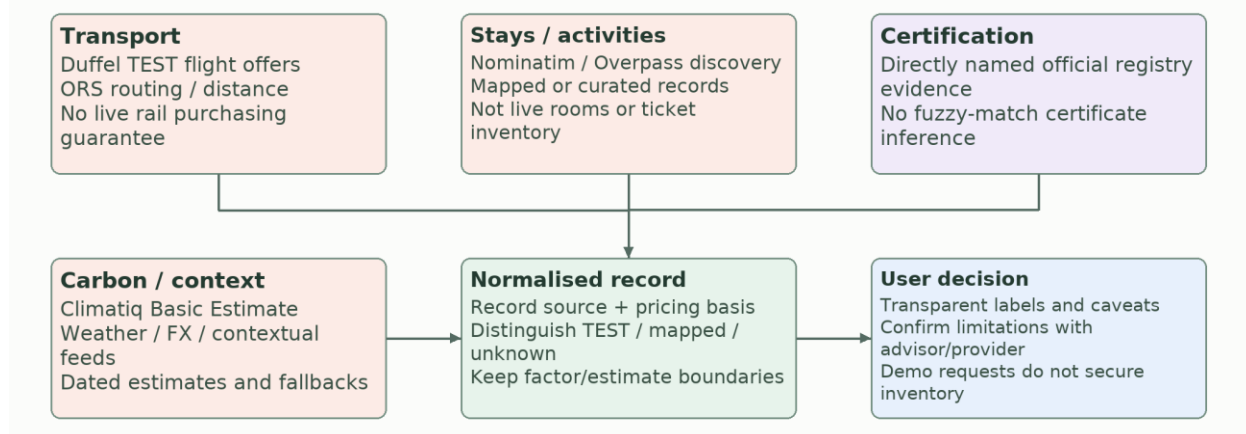


Figure 18. Provider inputs converging into labelled options with their limits

5.4 Weighted ranking

The options are scored within their own category, and each factor is minimum to maximum normalised within the category (lower carbon or price scores higher) and combined with the weights in Table 7 into a 0–100 score. Missing values then receive a neutral 0.5 rather than being treated as zero or free, and an option over budget in the same currency has its score multiplied by 0.35. For example, contributions of 0.8, 0.6, 0.7 and 0.5 under balanced weights give 0.685, displayed as 69. Every card shows the weights used (Figure 19), Listing 2 shows the transport version.

Priority	Carbon	Price	Preference	Convenience
Climate-first	60%	20%	10%	10%
Balanced (default)	45%	30%	10%	15%
Comfort-first	30%	20%	15%	35%

Table 7. Ranking weights by sustainability priority

Recommendation ranking and budget integrity

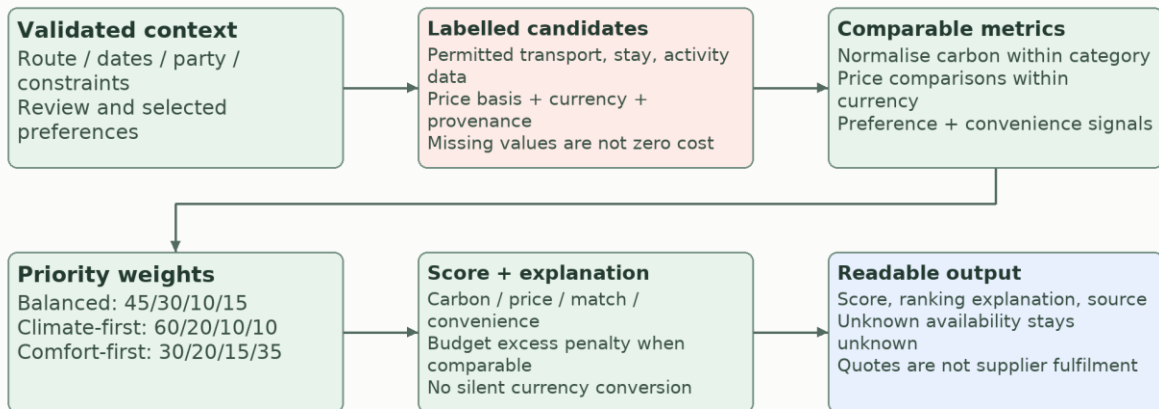


Figure 19. Category-local ranking with currency and quote-basis safeguards

```

487 def rank_options(
488     options: list[dict[str, Any]],
489     sustainability_level: str,
490     budget: Any = None,
491     transport_preference: Any = None,
492     travelers: int = 1,
493 ) -> list[dict[str, Any]]:
494     weights = WEIGHTS.get(sustainability_level, WEIGHTS["balanced"])
495     budget_limit = _budget_eur(budget)
496     preferred_mode = _mode_for_preference(transport_preference)
497     ranked = []
498     for option in options:
499         carbon_score = _normalise_inverse(float(option["carbon_kg"]), 350.0)
500         price = float(option["price_eur"]) * max(1, travelers)
501         if budget_limit is None:
502             price_score = _normalise_inverse(price, 500.0)
503             budget_score = None
504         else:
505             price_score = _normalise_inverse(price, budget_limit)
506             budget_score = round(max(0.0, 1.0 - max(0.0, price - budget_limit) / budget_limit), 3)
507         preference_score = (
508             0.5 if preferred_mode is None
509             else 1.0 if option.get("id") == preferred_mode
510             else 0.0
511         )
512         score = (
513             carbon_score * weights.carbon
514             + price_score * weights.price
515             + preference_score * weights.preference
516         )
517         if budget_score is not None:
518             score = score * 0.85 + budget_score * 0.15
519         ranked.append({
520             **option,
521             "preference_fit": preference_score,
522             "budget_fit": budget_score,
523             "score": round(score * 100, 1),
524         })
525     return sorted(ranked, key=lambda item: item["score"], reverse=True)

```

Listing 2. Transport ranking in rasa-bot/actions/actions.py

5.5 Handover packaging

When the traveller asks for a person, or the assistant cannot meet a constraint, the handover action packages the validated trip, budget, priority, access needs, selected options, unanswered questions and a bounded transcript. GPS coordinates, login tokens and unrelated history are excluded. Advisors use a role-protected queue with atomic assignment as shown on Figure 20 below.

Human-advisor handover: consent and ownership

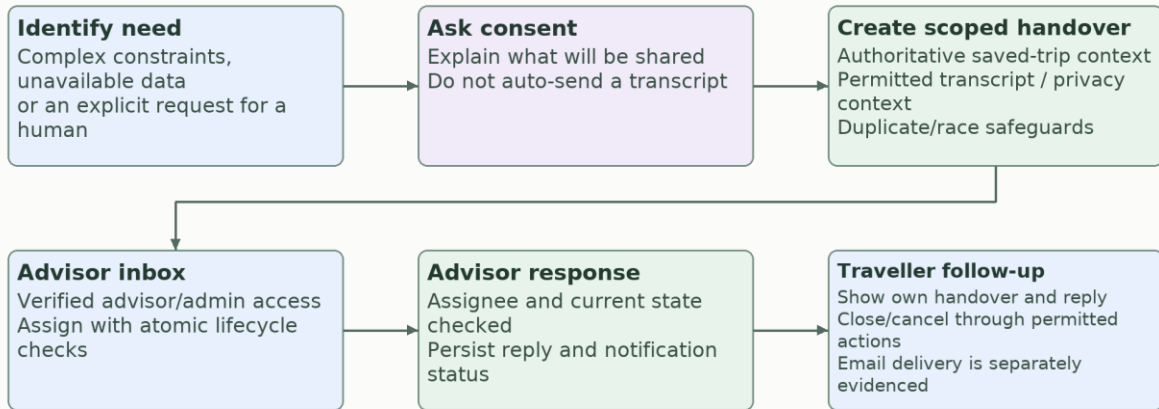


Figure 20. Handover creation, consent, assignment, reply and traveller follow-up

5.6 Persistence, privacy and access control

Data model: all 15 application tables

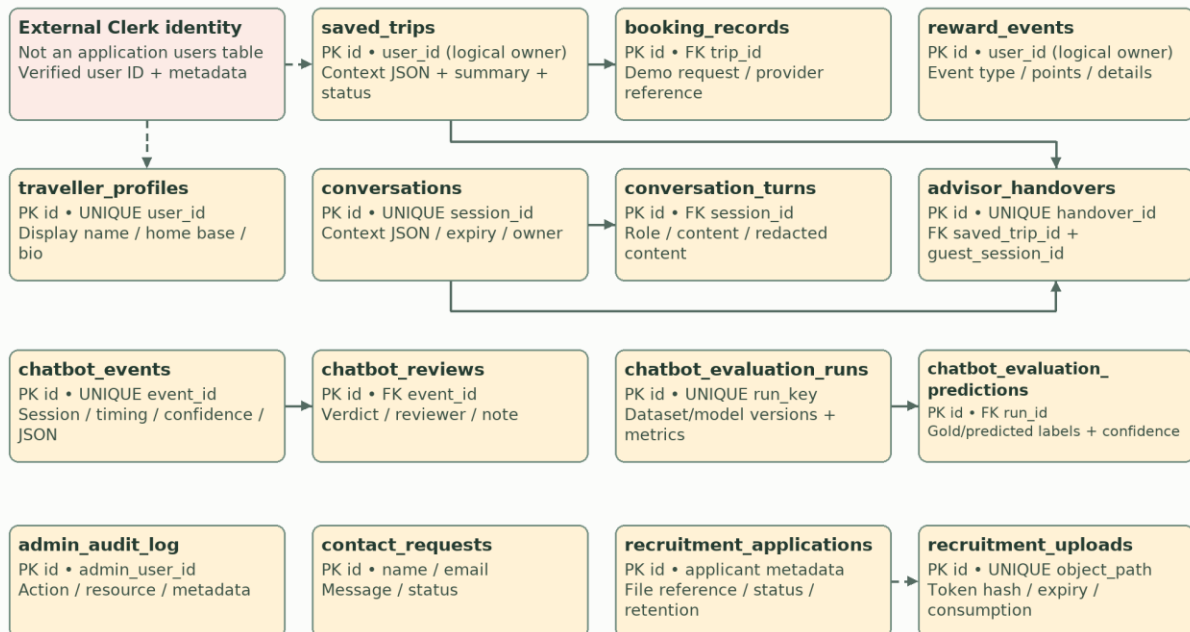


Figure 21. Entity relationship diagram of all 15 application tables

Owned-trip saving and resumption

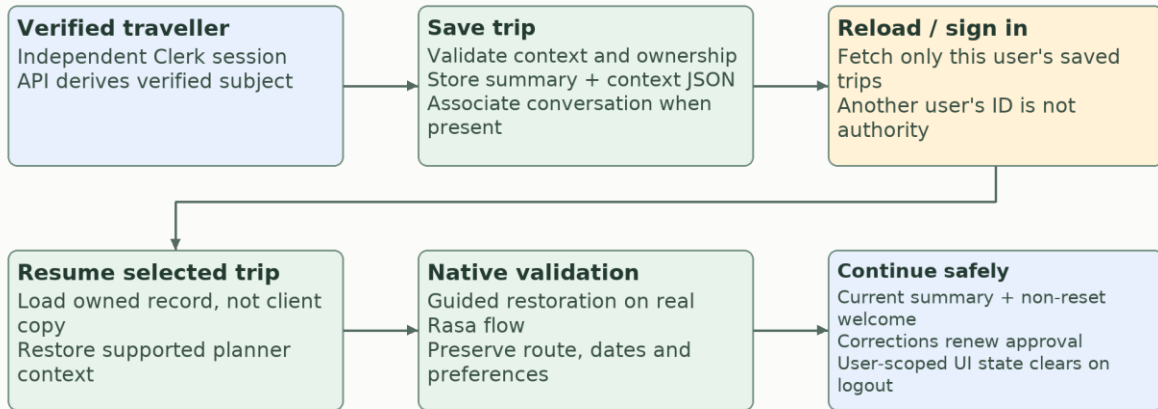


Figure 22. Saving, reloading and resuming an owned trip

Identity, conversation and file-data lifecycles

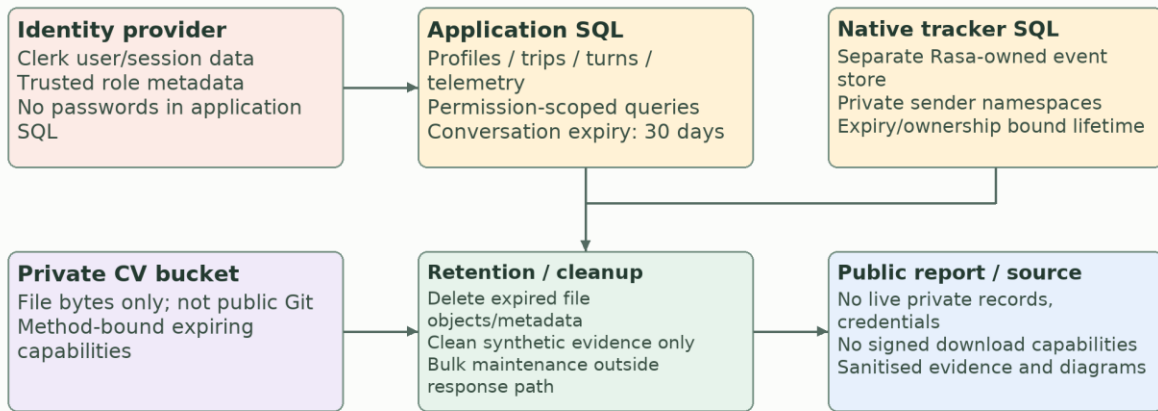


Figure 23. Identity, conversation and file-data lifecycles with separate retention

Authentication and mobile frame boundary

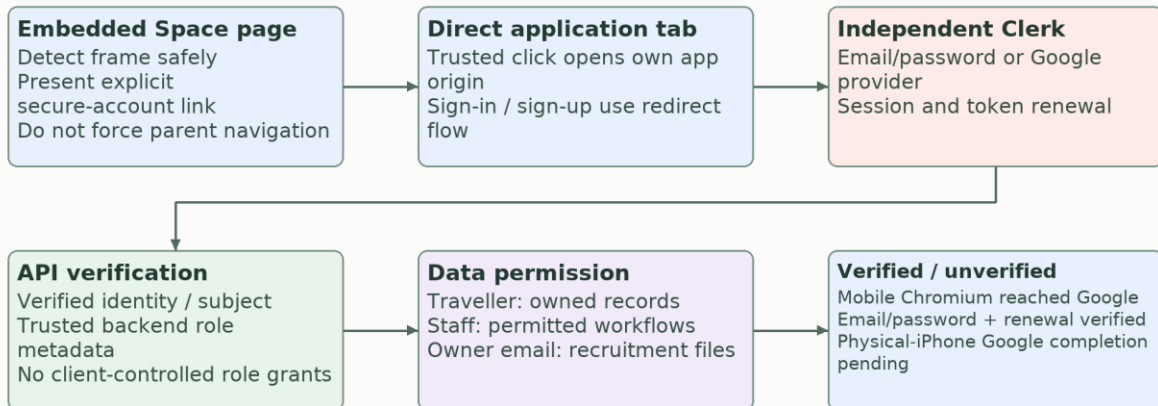


Figure 24. Authentication boundary: verified identity and API-enforced permissions

6. TESTING

6.1 Strategy

I then tested in layers because each layer answers a different question, for example NLU tests measure classification and Core tests measure action prediction while source tests check code, cross-validation checks how well the model generalises and finally user testing checks complete tasks end to end. Figure 25 shows these layers.

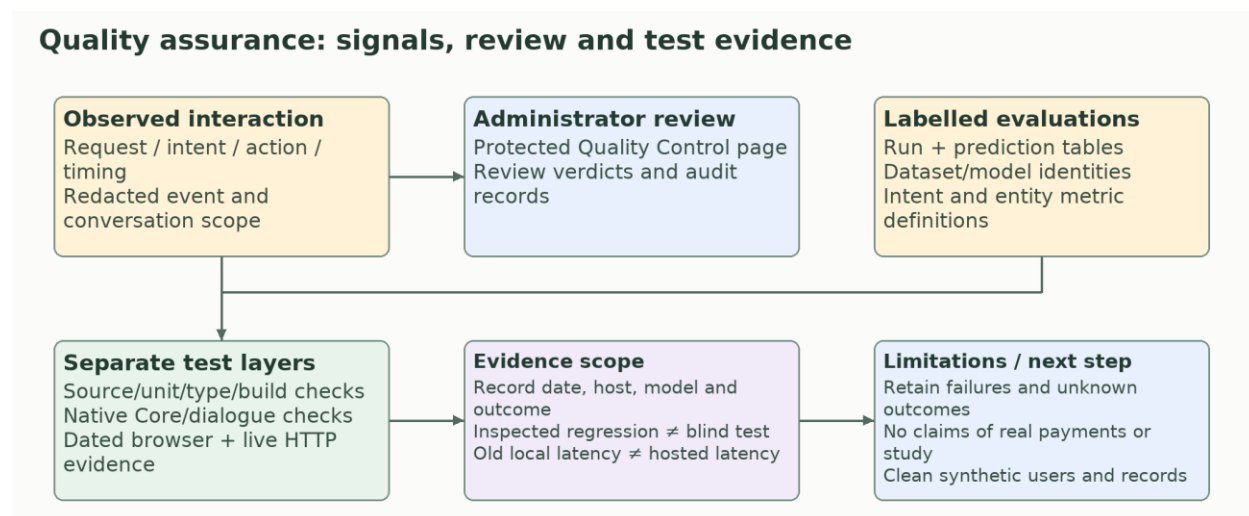


Figure 25. Quality assurance: interaction signals, review and separate test layers

6.2 NLU accuracy and cross-validation

I used a full-pipeline evaluator alongside rasa test nlu so the custom extractor was included in entity scores (Table 8).

Test set	Cases	Intent accuracy	Entity precision	Entity recall	Entity F1
Main labelled set	48	40/48 (83.3%)	1.000	0.750	0.857
Entity held-out set	26	18/26 (69.2%)	0.829	0.707	0.763
Generalisation set	48	Not labelled	0.767	0.749	0.758

Table 8. NLU results on the published model

The eight intent errors (Table 9) include close neighbours, such as a carbon-method question read as a carbon-footprint question, and short route statements sent to fallback, which is safe but wrong. The most concerning are access needs such as “wheelchair access” read as a location choice, because an access need could be lost. Precision of 1.000 with recall of 0.750 means the extractor is cautious and misses about a quarter of entities.

Utterance	Expected	Predicted
what calculation produced this carbon score	ask_carbon_method	ask_carbon_footprint
travel from Paris to Rome	inform	nlu_fallback
take me to Newport	inform	nlu_fallback
go to Rome via Milan	inform	remove_stopover
remove that stopover	correct_information	remove_stopover
wheelchair access	inform	choose_location_mode
I might visit Cambridge	inform	choose_location_mode
calculate the area of a triangle	out_of_scope	nlu_fallback

Table 9. Intent errors on the main labelled set

I then ran five-fold stratified cross-validation on all 496 training examples across 21 intents (seed 42). Each fold trained the same 100-epoch DIET configuration on four folds and tested on the fifth and in separate Rasa processes because the single-process command ran out of memory. Table 10 shows the results I got.

Fold	Test n	Accuracy	Macro P	Macro R	Macro F1	Weighted F1
1	100	67.00%	0.691	0.682	0.675	0.666
2	99	63.64%	0.569	0.568	0.549	0.597
3	99	70.71%	0.769	0.681	0.680	0.687
4	99	62.63%	0.586	0.628	0.576	0.605
5	99	68.69%	0.690	0.670	0.653	0.687
Pooled	496	66.53%	0.658	0.643	0.636	0.658

Table 10. Five-fold cross-validation intent metrics

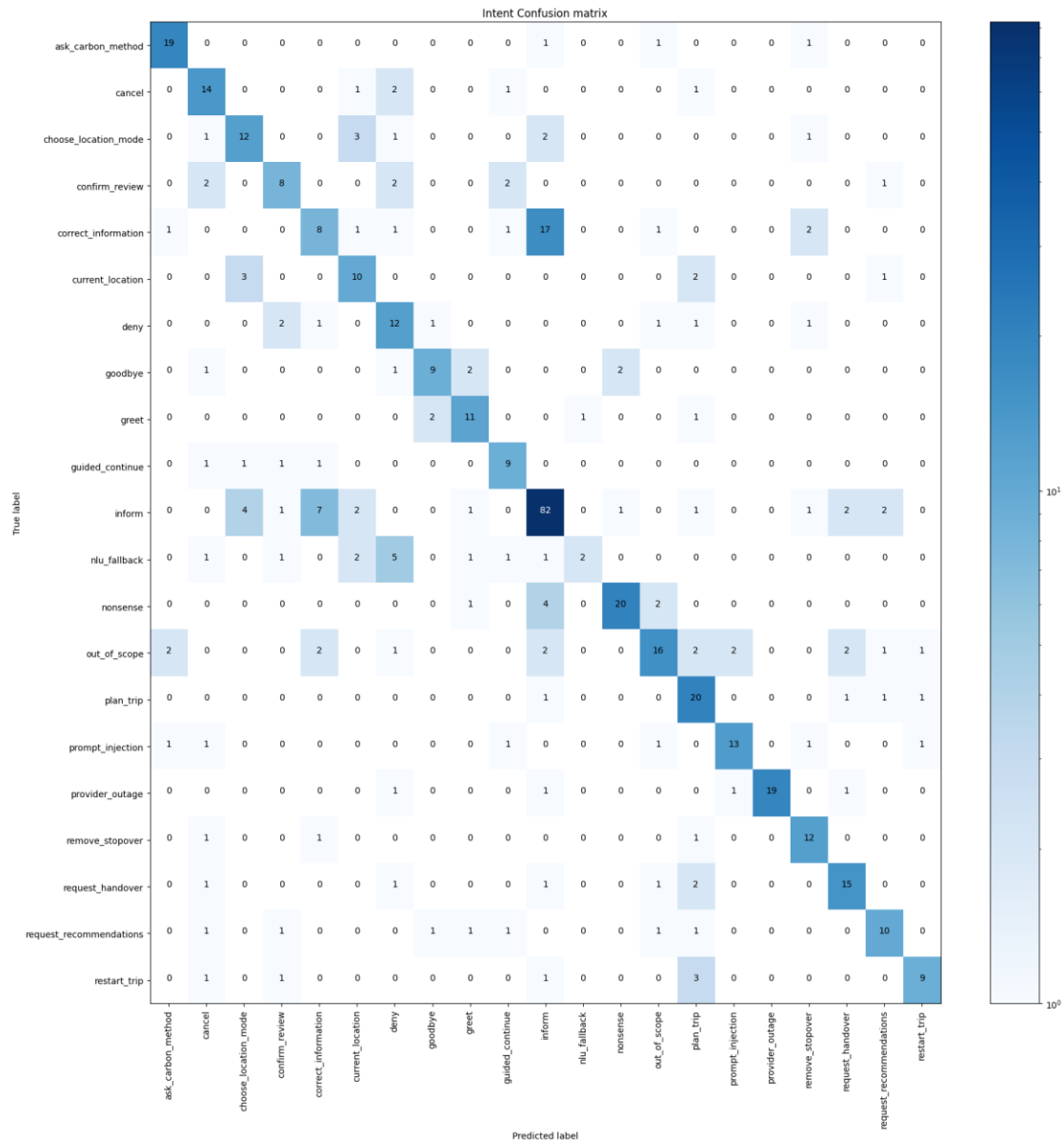


Figure 26. Pooled out-of-fold intent confusion matrix (496 predictions; actual on rows, predicted on columns)

6.3 Dialogue and source testing

Rasa test core passed all 16 test stories and 84 of 84 actions, covering happy paths, corrections, refusal, FAQ interruptions and handover. Fixed stories cannot prove every free-text conversation works, and browser testing found two failures for example “GBP 1,500 total budget for the trip” was rejected, and a question about the balanced priority received a generic answer. All 132 API, 22 frontend and 105 Python source tests however passed. In 29 timed turns every reply appeared within three seconds, but cold starts and load were not measured.

6.4 User testing

Due to time constraints, I first tested the system myself, and writing down the problems and improvement ideas as I went. I then asked one other person to complete the same tasks. Their participation was completely voluntary and their feedback was taken down with their consent. Table 11 summarises what we found.

Task	What we found	Recommendation
T1 Plan a Berlin-Paris trip	Trip details were collected, reviewed and compared correctly. Typing an exact budget such as “EUR 1,000 total budget for the whole trip” was not understood, so a budget band had to be chosen instead	Add exact-amount budget entry and accept more currency formats
T2 Change the origin to Leipzig	Changing only the departure city kept the destination and every other detail, and a new review was required before results	No change needed
T3 Explain carbon and certification	The carbon method was explained clearly (distance, mode, occupancy, Climatiq versus demo factors and offsets), but asking whether a mapped hotel is certified received a generic reply	Add an intent that answers certification questions from the registry status on the card
T4 Find and explain human help	The advisor option was easy to find and clearly stated that it is not live chat, but the chat did not list which trip details would be shared, and longer questions about consent were not understood	Explain the shared details and consent step in the chat as well as in the panel

Table 11. User testing findings and recommendations

7. DEPLOYMENT

The application runs as a single Docker image on Hugging Face Spaces (Figure 27). Nginx is the only public entry point on port 7860. It serves the React build and forwards API calls to Express, while Rasa (5005) and the action server (5055) listen only inside the container so a public request to the Rasa webhook returns 404.

Current Hugging Face deployment

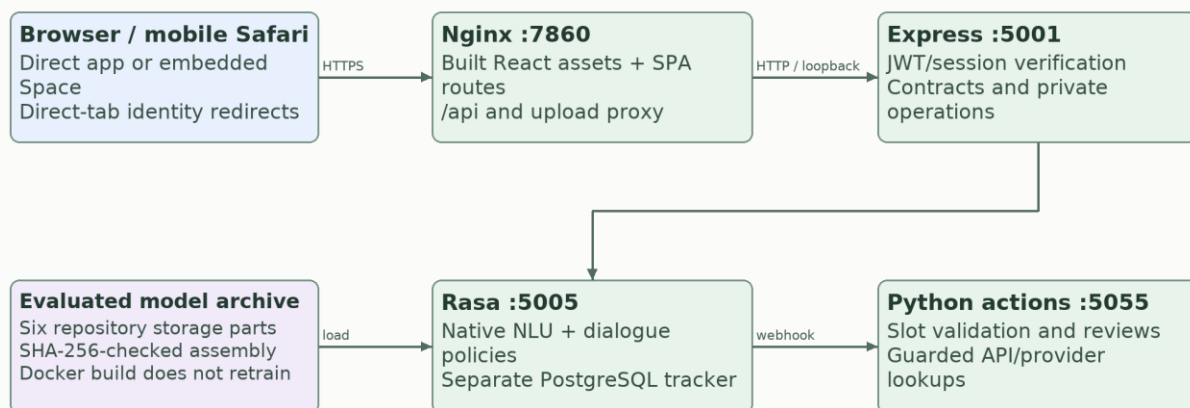


Figure 27. Hosted Docker topology on Hugging Face Spaces

Figure 28 shows the six deployment steps and the risk each one controls.

Deployment steps: from repository to live Hugging Face Space

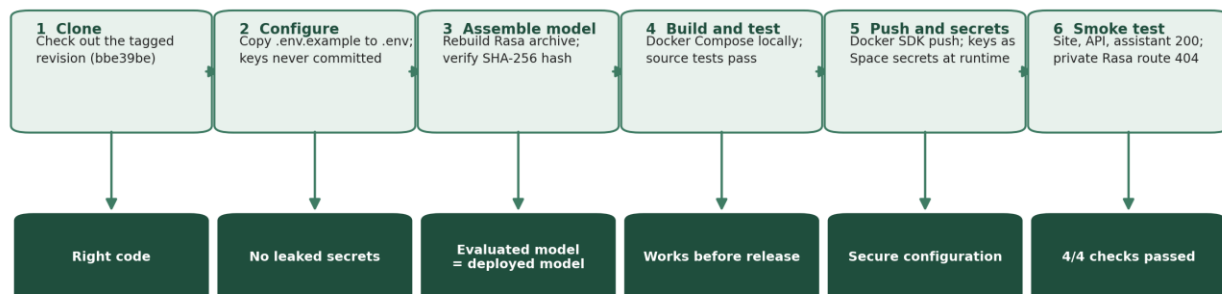


Figure 28. Deployment steps and the risk each one controls

Published release and assessment evidence

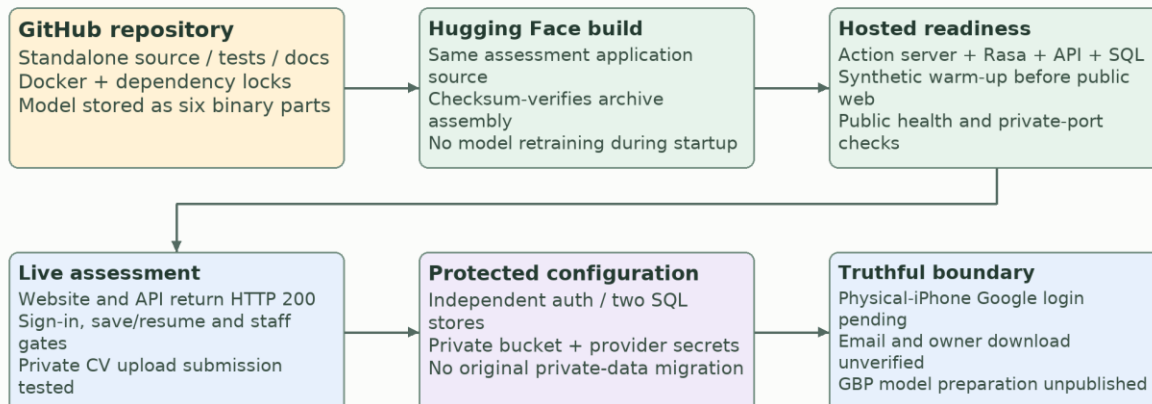


Figure 29. Published release evidence and its verified boundaries

8. CONCLUSION

In conclusion, for this project I was able to develop a functional platform for users to be able to get verified sustainable travel options ranging from transportation, accommodation and activity preferences. I also prioritised ensuring the sources are made clear and I was transparent on how I ranked the options I presented. Overall, the rasa powered assistant achieved the objectives I set and I included the functionality to have a human to be requested when the user wants to and with that I added consent and transparent mechanisms such as allowing the user to see what the human advisor receives from their end and for the user to consent.

The live application was then deployed on Hugging face. In the application for signed in user I added a rewards functionality and that will enable user to behave in more sustainable ways and for them to receive updates on how they are doing and to encourage them too. That functionality is still in development and went above the scope of this assignment. For future direction, I aim to launch and make this fully functional as well as add bookings on the platform and tracking of options selected so I can also measure the impact of the rasa assistant. On a technical reflection, building Landa travels showed me that being honest is harder than being helpful. All 16 dialogue stories and 259 source tests passed and changing the departure city worked every time in user testing, but the cross-validated intent accuracy of 66.5% showed that the model still relies on the phrasing it was trained on. User testing also showed that typed budgets and questions about hotel certification are not yet fully understood, and certified hotel data covers only a few cities. I will also conduct more user testing with travellers and improve the assistant and platform as a whole.

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APPENDIX A. PROJECT LINKS

Live application on Hugging Face Spaces: <https://yolandankala-landa-travels.hf.space>

GitHub repository : <https://github.com/Yolanda2020Code/landa-travels>

APPENDIX B. REPRODUCTION COMMANDS

Source checkout, model assembly and source tests:

```
git clone https://github.com/Yolanda2020Code/landa-travels.git
cd landa-travels
git checkout bbe39be9be5a50cd8ab48c2976201ff3b8d73806
pnpm install --frozen-lockfile
python deployment/assemble-model.py      # rebuilds model, checks SHA-256
pnpm build && pnpm test
cd rasa-bot && PYTHONPATH=. python -m pytest tests -q
```

NLU and dialogue evaluation, run from the rasa-bot directory:

```
PYTHONPATH=. python scripts/evaluate-entity-spans.py \
--model models/landa-travels.tar.gz \
--nlu tests/test_nlu.yml tests/entity_held_out.yml tests/entity_generalisation.yml \
--out report-evidence/current-nlu.json

rasa test core --model models/landa-travels.tar.gz \
--stories tests/test_stories.yml --out report-evidence/core --no-plot
```

Five-fold cross-validation (process-isolated runner; evaluation models are not deployed):

```
python reports/run_isolated_five_fold_cv.py
python reports/package_five_fold_cv.py
```

Hosted smoke check (expected: website 200, API 200, assistant 200, private Rasa route 404):

```
python deployment/smoke-huggingface.py https://yolandankala-landa-travels.hf.space
```

Environment: Python 3.10, Rasa 3.6.21, rasa-sdk 3.6.2, spaCy 3.7.5 with en_core_web_md 3.7.1, Node 22 and pnpm 10.26.1 and the published model SHA-256 is:

```
954da34118ed358fb80d8a2c60bca9081f78d2a6209c6dc8a6980482fa44b320
```

APPENDIX C. BRIEF REQUIREMENTS AND HOW THEY ARE MET

Brief element	Implementation	Match
Climatiq carbon per transport mode	Climatiq Basic Estimate in Python actions; unavailable state on failure	Full
Amadeus hotels and flights	OSM hotel data (course substitute); Duffel test flights; Amadeus self-service closed 17 July 2026	Substituted
Weighted carbon, price, preference ranking	Four-factor category-relative score, priority-specific weights	Full
Full context handover	Consented, bounded summary with trip, options and open questions	Full
DIET with spaCy	As Table 5; DistilBERT not used (optional)	Full
Stories, rules and slots	Form with validated slots; 16 test stories	Full
Two-stage fallback before handover	action_two_stage_fallback outside the form; form-level clarification inside (Section 5.2)	Partial
Dynamic quick replies	Rendered from assistant responses	Full
Green, amber, red results card	Coloured label with text and kg value on each option	Full
Human handover indicator	Status line with request reference after consent	Full
Optional voice and multiple languages	Browser voice input and read-aloud; English only	Partial

Table C1. Required elements of Task 4

APPENDIX D. USER TESTING PROTOCOL

I read and shared the following script to my one user for testing:

Please note this is user testing for the purposes of improving the development of my Landa Travels platform which I am developing for my Masters in Artificial Intelligence. Participation is completely voluntary and I can only proceed with consent, including taking notes. Please use the platform and mention outloud your observations and what you like or what you would like to be changed. I also have four tasks that I will ask you to do as you use the chatbot. Thank you for your participation.

Task	What the tester does	Success means	Cap
T1	Plan a Berlin–Paris trip for two adults, future dates, €1,000 budget, balanced priority; review and approve	Details kept; results or an explicit unavailable message	10 min
T2	Change only the departure city to Leipzig; review and approve again	Origin changes; everything else kept; old approval cancelled	5 min
T3	Explain where a carbon estimate comes from and one limitation; distinguish a mapped stay from a certified stay	Record the actual explanation	5 min
T4	Find human help and explain the summary, consent and asynchronous replies; do not submit	Found and understood	5 min

Table D1. Tasks

APPENDIX F. PLATFORM SCREENSHOTS

Screenshots of the live application on Hugging Face Spaces, in the order a traveller meets them.

F.1 Planning conversation

The screenshot displays the Landa Travels website interface. At the top, the logo "Landa Travels RESPONSIBLE JOURNEYS" is visible alongside navigation links: About, Research, Team, Contact, Planner, Sign in, and a button "Plan without signing in". Below the header, a banner reads "AI travel assistant · Carbon figures are estimates; check prices and availability before booking. [Privacy & data use](#)".

The main chat area shows a conversation with an AI assistant named Landa. The assistant's first message is: "Hello — I'm Landa, your AI travel-planning assistant, not a human advisor. I'll ask one question at a time; you can also type several details together. Carbon figures are estimates, not guarantees. Location sharing is optional. Avoid sensitive details or an exact home address. You can request an advisor anytime." The user's response is: "Where will you travel from? Please give a city or region, not an exact address." Below this, a form titled "Where will you travel from?" is shown with a label "ORIGIN" and a text input field containing "City or airport". A "Continue" button is at the bottom of the form.

On the right side, a "Trip Summary" panel is visible, listing various travel details: ROUTE (Not set), DATES (Not set), TRAVELLERS (1 total (1 adults, 0 children)), BUDGET (Not set), TRANSPORT (Not set), ACCOMMODATION (Not set), and ACCESSIBILITY (Not set). Each item has an edit icon. A "SESSION ONLY" button is at the top of the summary panel.

At the bottom of the chat area, there is a "Request advisor help" button, a "Listen" button, and a text input field with the placeholder "Type your answer or ask a question...". A small note states "Advisor requests are reviewed during staffed hours."

Figure F1. Opening message with AI disclosure and the origin question



Landa Travels
RESPONSIBLE JOURNEYS

[About](#)
[Research](#)
[Team](#)
[Contact](#)
[Planner](#)
[Sign In](#)

Plan without signing in

AI travel assistant · Carbon figures are estimates; check prices and availability before booking. [Privacy & data use](#)



Hello — I'm Landa, your AI travel-planning assistant, not a human advisor. I'll ask one question at a time; you can also type several details together. Carbon figures are estimates, not guarantees. Location sharing is optional. Avoid sensitive details or an exact home address. You can request an advisor anytime.



Where will you travel from? Please give a city or region, not an exact address.

My origin is Berlin (BER).



Where would you like to go from Berlin (BER)? Choose a configured city or type another city. For a recognized origin, examples are ordered by straight-line proximity:

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

This weekend

Next month

Dates are flexible

Type your answer or ask a question...




Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

Not set

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

Not set

TRANSPORT

Not set

ACCOMMODATION

Not set

ACCESSIBILITY

Not set

Request advisor help

Figure F2. Origin given in free text; the trip summary updates



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RESPONSIBLE JOURNEYS

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[Contact](#)
[Planner](#)
[Sign In](#)

Plan without signing in

AI travel assistant · Carbon figures are estimates; check prices and availability before booking. [Privacy & data use](#)



exact home address. You can request an advisor anytime.



Where will you travel from? Please give a city or region, not an exact address.

My origin is Berlin (BER).



Where would you like to go from Berlin (BER)? Choose a configured city or type another city. For a recognized origin, examples are ordered by straight-line proximity; this is not a transport route, schedule, travel time, or availability.

Paris

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

This weekend

Next month

Dates are flexible

Type your answer or ask a question...




Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

Not set

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

Not set

TRANSPORT

Not set

ACCOMMODATION

Not set

ACCESSIBILITY

Not set

Request advisor help

Figure F3. Destination question and answer

The screenshot shows the Landa Travels website's AI travel assistant. The header includes the logo, navigation links (About, Research, Team, Contact, Planner), and a 'Sign In' button. A 'Plan without signing in' button is also present. The main chat area displays a conversation where the user asks, 'When are you planning to travel? Approximate dates are fine.' The assistant responds with a 'Paris' button and a 'Change origin to Leipzig' button. Below this, there's a section for 'Choose your travel dates' with options for 'This weekend', 'Next month', and 'Dates are flexible'. A text input field at the bottom allows the user to 'Type your answer or ask a question...'. On the right, a 'Trip Summary' sidebar lists details: ROUTE (Leipzig → Paris), DATES (Not set), TRAVELLERS (1 total (1 adults, 0 children)), BUDGET (Not set), TRANSPORT (Not set), ACCOMMODATION (Not set), and ACCESSIBILITY (Not set). A 'Request advisor help' button is at the bottom right.

Figure F4. Changing only the origin to Leipzig mid-conversation

This screenshot shows the same Landa Travels AI travel assistant interface, but with a more advanced date picker. The user's question is the same. The assistant's response now includes a 'Choose your travel dates' section with 'DEPARTURE' (5. Oct 2026) and 'RETURN' (11. Oct 2026) fields. A 'My return date is flexible' button and a 'Continue' button are also visible. The 'Trip Summary' sidebar on the right remains the same, showing the route as Leipzig → Paris and other travel details. The 'Request advisor help' button is still at the bottom right.

Figure F5. Date picker with quick replies

Landa Travels

RESPONSIBLE JOURNEYS

About

Research

Team

Contact

Planner

Sign In

Plan without signing in

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What total budget and currency should I work within?

What total budget should I work within?

Under €1,000

€1,000–2,500

€2,500–5,000

€5,000+

Continue

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Under €1,000

€1,000–2,500

Flexible budget

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026–10–05 to 2026–10–...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

Not set

TRANSPORT

Not set

ACCOMMODATION

Not set

ACCESSIBILITY

Not set

Request advisor help



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Which transport do you prefer or need for Leipzig to Paris? These choices guide indicative comparisons, not live availability.

How do you prefer to travel?



Train

Lower impact, scenic



Bus / Coach

Cost effective



Flight

For long distances

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Advisor requests are reviewed during staffed hours.

Rail

Coach

Flight

Flexible

Type your answer or ask a question...




Trip Summary

SESSION ONLY



ROUTE

Leipzig → Paris



DATES

2026-10-05 to 2026-10-...



TRAVELLERS

1 total (1 adults, 0 children)



BUDGET

€1,000-2,500



TRANSPORT

Not set



ACCOMMODATION

Not set



ACCESSIBILITY

Not set

[Request advisor help](#)



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How do you prefer to travel?



Train
Lower impact, scenic



Bus / Coach
Cost effective



Flight
For long distances



Compare all
Show practical alternatives

Continue

 Request advisor help
 Listen

Advisor requests are reviewed during staffed hours.

Rail

Coach

Flight

Flexible




Trip Summary
[SESSION ONLY](#)



ROUTE
Leipzig → Paris



DATES
2026-10-05 to 2026-10-...



TRAVELLERS
1 total (1 adults, 0 children)



BUDGET
€1,000-2,500



TRANSPORT
Not set



ACCOMMODATION
Not set



ACCESSIBILITY
Not set

 Request advisor help

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 Do you have accessibility or assistance needs, or should I record none?

Any accessibility requirements?

 **No specific needs**
Standard options are fine

 **Wheelchair access**
Step-free required

 **Limited mobility**
Minimise walking/stairs

 Request advisor help

 Listen

None

Step-free access

Wheelchair access

Mobility assistance

Type your answer or ask a question...

Trip Summary

SESSION ONLY

 ROUTE
Leipzig → Paris

 DATES
2026-10-05 to 2026-10-...

 TRAVELLERS
1 total (1 adults, 0 children)

 BUDGET
€1,000-2,500

 TRANSPORT
flight

 ACCOMMODATION
Not set

 ACCESSIBILITY
Not set

 Request advisor help



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How important is minimizing carbon impact?



Climate-first
Prioritise lower-carbon options



Balanced
Good eco-choices within reason



Comfort-first
Prioritise convenience and comfort

Continue

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Climate-first

Balanced

Comfort-first

Type your answer or ask a question...




Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

Not set

ACCESSIBILITY

none

Request advisor help

Figure F10. Sustainability priority: climate-first, balanced or comfort-first



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Where would you like to stay?



Eco-certified Hotel
Explore sourced certification claims



Local Guesthouse
Support local economy



Apartment
Self-catering

Continue

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Hotel

Hostel

Apartment

None

Type your answer or ask a question...




Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

Not set

ACCESSIBILITY

none

Request advisor help

Figure F11. Accommodation preference



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What activities would you enjoy?

Choose one focus for the recommendations.



Cultural experiences
Use this to tailor activity suggestions



Outdoor activities
Use this to tailor activity suggestions



Nature and wildlife
Use this to tailor activity suggestions



No preference
Use this to tailor activity suggestions

Continue

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Cultural experiences

Outdoor activities

Nature and wildlife

No preference

Type your answer or ask a question...




Trip Summary

SESSION ONLY



ROUTE

Leipzig → Paris





DATES

2026-10-05 to 2026-10-...





TRAVELLERS

1 total (1 adults, 0 children)





BUDGET

€1,000-2,500





TRANSPORT

flight





ACCOMMODATION

eco-hotel





ACCESSIBILITY

none



Request advisor help

Figure F12. Activity focus



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Would you like to use approximate device location, enter a city manually, or skip location sharing?

Location preferences
 Optional: use a city or approximate device location to find nearby options. Device coordinates are rounded to two decimals, kept only in this browser session, and sent only with the nearby-options query after you confirm your trip; they are not saved with your trip or sent to the Rasa assistant.



Allow approximate device location

Enter an approximate city

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Enter a city manually

Skip location

Type your answer or ask a question...




Trip Summary

SESSION ONLY



ROUTE

Leipzig → Paris





DATES

2026-10-05 to 2026-10-...





TRAVELLERS

1 total (1 adults, 0 children)





BUDGET

€1,000-2,500





TRANSPORT

flight





ACCOMMODATION

eco-hotel





ACCESSIBILITY

none



Request advisor help

Figure F13. Optional location sharing with privacy explanation

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F.2 Review and approval

The screenshot shows the Landa Travels AI travel assistant interface. The header includes the Landa Travels logo, navigation links (About, Research, Team, Contact, Planner), and a 'Sign In' button. A 'Plan without signing in' button is also present. A banner at the top states: 'AI travel assistant - Carbon figures are estimates; check prices and availability before booking. [Privacy & data use](#)'.

The main content area is divided into two sections. The left section contains a text review of collected details: 'Review: origin=Leipzig; destination=Paris; stopovers=none; dates=2026-10-05 to 2026-10-11; travellers=1; budget=€1,000-2,500; transport=flight; accessibility=none; sustainability=balanced; accommodation=eco-hotel; activities=nature; location_mode=gps. Confirm these details before I search.' Below this is a confirmation prompt: 'Are you happy with these details? Please confirm before I calculate your footprint and compare options.' A table lists the details: Route (Leipzig → Paris), Dates (2026-10-05 to 2026-10-11), Travellers (1 total (1 adults, 0 children)), Budget (€1,000-2,500), Transport (flight), Accommodation (eco-hotel), and Accessibility (none). Below the table are buttons for 'Confirm and compare options' and 'I want to change something', and a text input field for 'Type your answer or ask a question...'. A 'Request advisor help' button is also present.

The right section is titled 'Trip Summary' and includes a 'SESSION ONLY' button. It lists the trip details: ROUTE (Leipzig → Paris), DATES (2026-10-05 to 2026-10-11), TRAVELLERS (1 total (1 adults, 0 children)), BUDGET (€1,000-2,500), TRANSPORT (flight), ACCOMMODATION (eco-hotel), and ACCESSIBILITY (none). A 'Request advisor help' button is at the bottom.

Figure F14. Text review of every collected detail before searching

The screenshot shows the Landa Travels AI travel assistant interface. The header is identical to Figure F14. A banner at the top states: 'AI travel assistant - Carbon figures are estimates; check prices and availability before booking. [Privacy & data use](#)'.

The main content area is divided into two sections. The left section contains a confirmation prompt: 'Are you happy with these details? Please confirm before I calculate your footprint and compare options.' Below this is a table listing the trip details: Route (Leipzig → Paris), Dates (2026-10-05 to 2026-10-11), Travellers (1 total), Budget (€1,000-2,500), Transport (flight), Stay (eco-hotel), Accessibility (none), and Priority (balanced). Below the table are buttons for 'Confirm and compare options' and 'I want to change something', and a text input field for 'Type your answer or ask a question...'. A 'Request advisor help' button is also present.

The right section is titled 'Trip Summary' and includes a 'SESSION ONLY' button. It lists the trip details: ROUTE (Leipzig → Paris), DATES (2026-10-05 to 2026-10-11), TRAVELLERS (1 total (1 adults, 0 children)), BUDGET (€1,000-2,500), TRANSPORT (flight), ACCOMMODATION (eco-hotel), and ACCESSIBILITY (none). A 'Request advisor help' button is at the bottom.

Figure F15. Review card asking for explicit confirmation

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Route

Leipzig → Paris

Dates

2026-10-05 to 2026-10-11

Travellers

1 total

Budget

€1,000-2,500

Transport

flight

Stay

eco-hotel

Accessibility

none

Priority

balanced

Location

Approximate device location

No, change something

Yes, show my results

Request advisor help

Listen

Confirm and compare options

I want to change something

Type your answer or ask a question...

Advisor requests are reviewed during staffed hours.

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F16. Approve or change buttons on the review card

F.3 Results and carbon labels

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Confirm and compare options

Thanks — I will use only these details to compare options.

ESTIMATED FOOTPRINT

approximately 196.9 kg CO₂e for 1 traveller.

Estimated breakdown: Direct economy flight 127.9 kg, 6 nights from your selected dates 60.0 kg, and local transport 9.0 kg.

I ranked the options for Paris using your balanced priority. These are planning

Request advisor help

Listen

Type your answer or ask a question...

Advisor requests are reviewed during staffed hours.

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F17. Estimated trip footprint and its breakdown

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I ranked the options for Paris using your balanced priority. These are planning estimates, not guarantees. They use indicative route, occupancy, accommodation, and local-transport factors (illustrative 500 km route because this city pair was not geocoded); transport cards identify whether their carbon estimate came from Climatiq or a demo factor. Demo factors are not live provider data. Confirm live prices, availability, and certification before booking. The total footprint remains an indicative mixed-source planning estimate; a climate contribution does not reduce or erase that displayed total.



Results are Duffel TEST MODE offers with route-average Climatiq flight estimates or country-average hotel-night estimates. They are not bookings or certified eco-accommodation.



Registry evidence reflects the latest fresh official snapshot and is not an endorsement, booking offer, or guarantee of availability. Confirm current certification

 Request advisor help
  Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...
 


Trip Summary SESSION ONLY

- ROUTE Leipzig → Paris
- DATES 2026-10-05 to 2026-10-...
- TRAVELLERS 1 total (1 adults, 0 children)
- BUDGET €1,000-2,500
- TRANSPORT flight
- ACCOMMODATION eco-hotel
- ACCESSIBILITY none

 Request advisor help

Figure F18. Explanation of how options were ranked and what the estimates mean


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Registry evidence reflects the latest fresh official snapshot and is not an endorsement, booking offer, or guarantee of availability. Confirm current certification with the European Commission before booking. Any displayed stay carbon figure is an indicative country-average estimate, not a property-specific measurement. Prices and availability are not supplied by this registry.

No validated live or sandbox inventory was returned for a required accommodation or transport preference; an advisor can check availability.

required_inventory_unavailable



Stay options: scroll horizontally or use the navigation buttons




STAY

☐ Include in advisor request

Carbon not estim

STAY

☐ Incl

BOB

Chouchou

 Request advisor help
  Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...
 


Trip Summary SESSION ONLY

- ROUTE Leipzig → Paris
- DATES 2026-10-05 to 2026-10-...
- TRAVELLERS 1 total (1 adults, 0 children)
- BUDGET €1,000-2,500
- TRANSPORT flight
- ACCOMMODATION eco-hotel
- ACCESSIBILITY none

 Request advisor help

Figure F19. Unavailable inventory stated honestly, with an option to ask an advisor

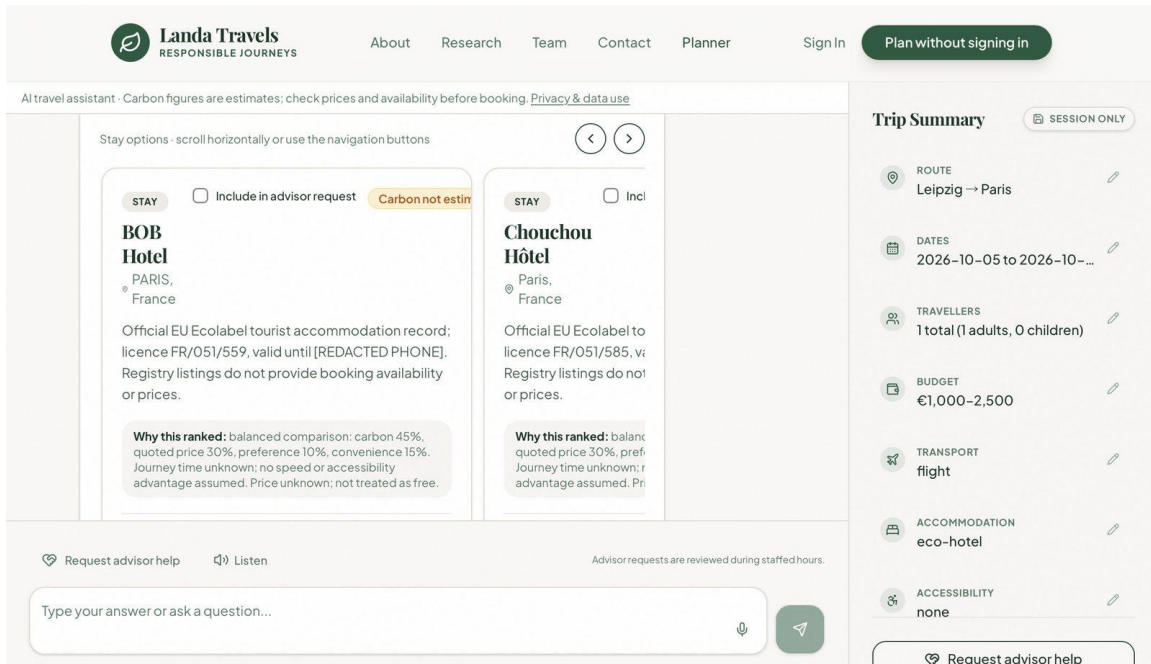


Figure F20. Stay carousel with EU Ecolabel records

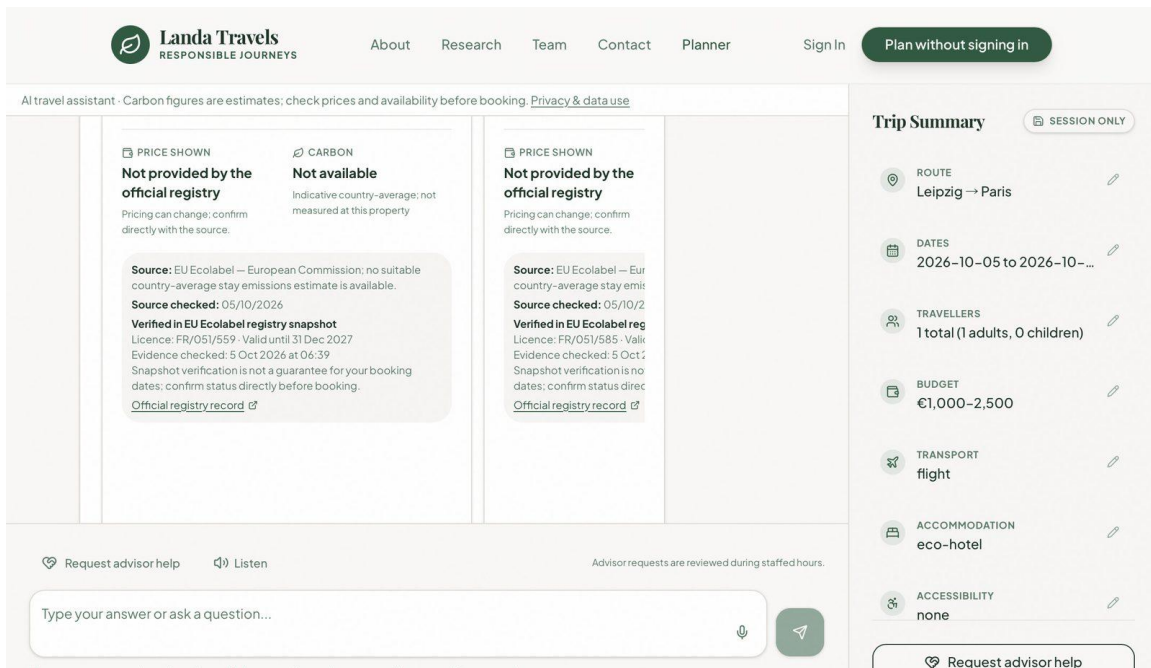


Figure F21. Stay card with registry licence and evidence date



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TRANSPORT

☐ Include in advisor request

low impact

Daytime rail

Paris

Approximate return journey over 1180 km. Indicative fare is per traveller; budget ranking considers the whole party. Confirm the actual route, timetable, fare, and occupancy.

Why this ranked:

balanced comparison: carbon 45%, quoted price 30%, preference 10%, convenience 15%. Journey time unknown; no speed or accessibility advantage assumed. Quoted component price, not the total itinerary cost.

PRICE SHOWN

€142 per traveller · indicative

CARBON

13.4 kg CO₂e

Pricing can change; confirm directly with the source. Estimate basis not specified

Source:

Climatiq estimate (GB conversion-factor proxy; passenger_train-route_type_international_rail-fuel_source_na); indicative mode estimate over 1180 km using

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F22. Daytime rail card: low impact with ranking explanation and Climatiq source



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TRANSPORT

☐ Include in advisor request

low impact

Direct coach

Paris

Approximate return journey over 1150 km. Indicative fare is per traveller; budget ranking considers the whole party. Confirm the actual route, timetable, fare, and occupancy.

Why this ranked:

balanced comparison: carbon 45%, quoted price 30%, preference 10%, convenience 15%. Journey time unknown; no speed or accessibility advantage assumed. Quoted component price, not the total itinerary cost.

PRICE SHOWN

€75 per traveller · indicative

CARBON

45.4 kg CO₂e

Pricing can change; confirm directly with the source. Estimate basis not specified

Source:

Climatiq estimate (GB conversion-factor proxy; passenger_vehicle-vehicle_type_coach-fuel_source_na-distance_na-engine_size_na); indicative mode estimate over 1150 km using illustrative 500 km route because this city pair was not geocoded: illustrative

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F23. Direct coach card: low impact



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EXPERIENCE

☐ Include in advisor request

low impact

Nature and wildlife day near Paris

Paris

A nature-oriented day using local public transport and walking where practical. Site conditions and availability must be confirmed.

Why this ranked: balanced comparison; carbon 45%, quoted price 30%, preference 10%, convenience 15%. Journey time unknown; no speed or accessibility advantage assumed. Quoted component price, not the total itinerary cost.

PRICE SHOWN

€24 per traveller · indicative
Pricing can change; confirm directly with the source.

CARBON

9 kg CO₂e
Estimate basis not specified

Source: Curated demonstration estimate

Source checked: 05/10/2026

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.




Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F24. Experience card: nature day near Paris



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PRICE SHOWN

€5 per traveller · indicative
Pricing can change; confirm directly with the source.

CARBON

196.9 kg CO₂e
Estimate basis not specified

Source: Curated guidance — programme and retirement record must be verified

Source checked: 05/10/2026

Unverified certification claim: Look for Gold Standard or Verra registry evidence; verify independently

No current official EU Ecolabel evidence is attached to this recommendation.

TRANSPORT

☐ Include in advisor request

moderate impact

Duffel Airways

Leipzig/Halle Airport → Paris

Approx. 190 min

Duffel TEST MODE return offer. Route-average Climatiq estimate is not aircraft-specific. Price and availability must be revalidated before any booking; Landa never creates an order here.

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.




Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F25. Flight card: Duffel test-mode offer with moderate impact

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specific. Price and availability must be revalidated before any booking; Landa never creates an order here.

Why this ranked: balanced comparison: carbon 45%, quoted price 30%, preference 10%, convenience 15%. Provider journey time 190 minutes, 0 connections; excludes door-to-door travel. Quoted component price, not the total itinerary cost.

PRICE SHOWN

EUR 82.09

Pricing can change: confirm directly with the source.

CARBON

191.2686517896 kg CO₂e

Estimate basis not specified

Source: Duffel TEST MODE + ClimaTiq outbound and return route estimates

Source checked: 03/10/2026

AROUND YOUR DESTINATION

Paris

Trip dates: 2026-10-05 to 2026-10-11

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F26. Flight card detail: weights, price shown, carbon and source

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OFFSET

☐ Include in advisor request

high impact

Verified climate contribution

Paris

Only after reducing the trip footprint, consider a programme that publishes its methodology, additionality evidence, registry serial numbers, and retirement records. The displayed 196.9 kg CO₂e is the footprint to address, not emissions erased by payment.

Why this ranked: balanced comparison: carbon 45%, quoted price 30%, preference 10%, convenience 15%. Journey time unknown; no speed or accessibility advantage assumed. Quoted component price, not the total itinerary cost.

High-carbon option

This option has a high estimated impact: 196.9 kg CO₂e. Consider a lower-carbon alternative if practical.

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE

Leipzig → Paris

DATES

2026-10-05 to 2026-10-...

TRAVELLERS

1 total (1 adults, 0 children)

BUDGET

€1,000-2,500

TRANSPORT

flight

ACCOMMODATION

eco-hotel

ACCESSIBILITY

none

Request advisor help

Figure F27. Offset card: high impact with the high-carbon warning

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F.4 Certified stays and mapped places

The screenshot displays the Landa Travels website interface. The header includes the logo, navigation links (About, Research, Team, Contact, Planner), and a 'Sign In' button. A 'Plan without signing in' button is also present. Below the header, a banner reads 'AI travel assistant - Carbon figures are estimates; check prices and availability before booking. Privacy & data use'.

The main content area is divided into two columns. The left column features a section titled 'Official EU Ecolabel stays' with a 'Registry checked' status. It includes a disclaimer about registry evidence and a verification note. Below this, two hotel listings are shown:

- BOB Hotel** (EU Ecolabel - verified in registry snapshot): PERNETY, PARIS, France. Scheme: EU Ecolabel. Licence: FR/051/559. Valid until: 31 Dec 2027. Evidence checked: 5 Oct 2026 at 06:39. Links: Official registry record, Hotel website.
- Chouchou Hôtel** (EU Ecolabel - verified in registry snapshot): Rue du Helder, Paris, France.

The right column displays a 'Trip Summary' for a session only, including details on route (Leipzig → Paris), dates (2026-10-05 to 2026-10-...), travellers (1 total), budget (€1,000-2,500), transport (flight), accommodation (eco-hotel), and accessibility (none). A 'Request advisor help' button is at the bottom.

At the bottom of the main content area, there is a chat interface with a 'Request advisor help' button, a 'Listen' icon, and a text input field for user queries.

Figure F28. Official EU Ecolabel stays with registry evidence

This screenshot shows the Landa Travels website with a different set of hotel listings. The header and navigation elements are identical to Figure F28. The 'Official EU Ecolabel stays' section lists three hotels:

- DREAM HOTEL OPERA** (EU Ecolabel - verified in registry snapshot): Rue le Peletier, PARIS, France. Scheme: EU Ecolabel. Licence: FR/051/591. Valid until: 31 Dec 2027. Evidence checked: 5 Oct 2026 at 06:39. Links: Official registry record, Hotel website.
- Grand Hotel Chicago** (EU Ecolabel - verified in registry snapshot): RUE DE ROME, PARIS, France. Scheme: EU Ecolabel. Licence: FR/051/593. Valid until: 31 Dec 2027. Evidence checked: 5 Oct 2026 at 06:39. Links: Official registry record, Hotel website.
- HOTEL AMELIE** (EU Ecolabel - verified in registry snapshot): rue Amélie, PARIS, France.

The 'Trip Summary' on the right remains the same as in Figure F28. The chat interface at the bottom is also present.

Figure F29. Further EU Ecolabel hotels with licence numbers and validity



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OpenStreetMap places

Map snapshot available

Mapped hotels (certification not verified) (20)

OpenStreetMap locations only. These map listings are not matched to the official certification records above. Live hotel inventory, prices and availability are unavailable without Stays provider access.

Ibis
 48.8403, 2.3156 · OSM node/224711659
[Open map record](#)

Hôtel Saint-Honoré
 48.8612, 2.3436 · OSM node/247435032
[Open map record](#)

Hôtel Montpensier
 48.8644, 2.3363 · OSM node/247435275
[Open map record](#)

Request advisor help
 Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE
 Leipzig → Paris

DATES
 2026-10-05 to 2026-10-...

TRAVELLERS
 1 total (1 adults, 0 children)

BUDGET
 €1,000-2,500

TRANSPORT
 flight

ACCOMMODATION
 eco-hotel

ACCESSIBILITY
 none

Request advisor help

Figure F30. Mapped hotels from OpenStreetMap, marked as not certified



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Nearby mapped places

Approximate GPS

Approximate GPS area near Berlin
 Nearby places are approximate straight-line distances from the rounded device location. They come from an offline OpenStreetMap snapshot checked 2026-09-30T11:01:41Z; map points are not live hotel inventory, transit timetables, route directions, or opening-hour information. GPS coordinates are not stored or included in this response.

Mapped hotels (6)

Mapped transit stops (6)

Mapped attractions (6)

Hotel ibis Berlin Mitte
 ≈ 1.5 km · [OSM source](#)

Bob W.
 ≈ 2.7 km · [OSM source](#)

Radisson Collection Hotel Berlin
 ≈ 2.8 km · [OSM source](#)

art'otel Berlin Mitte
 ≈ 2.8 km · [OSM source](#)

Prenzlauer Allee/Danziger Straße
 ≈ 1.5 km · [OSM source](#)

Prenzlauer Allee/Danziger Straße
 ≈ 1.5 km · [OSM source](#)

Storkower Straße
 ≈ 1.8 km · [OSM source](#)

S Prenzlauer Allee
 ≈ 1.9 km · [OSM source](#)

Jugend[widerstands]museum
 ≈ 1.7 km · [OSM source](#)

Galgenhaus Stadtmuseum Berlin
 ≈ 3.1 km · [OSM source](#)

Hugenottenmuseum Berlin
 ≈ 3.7 km · [OSM source](#)

Parlamentshistorische

Request advisor help
 Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

Trip Summary

SESSION ONLY

ROUTE
 Leipzig → Paris

DATES
 2026-10-05 to 2026-10-...

TRAVELLERS
 1 total (1 adults, 0 children)

BUDGET
 €1,000-2,500

TRANSPORT
 flight

ACCOMMODATION
 eco-hotel

ACCESSIBILITY
 none

Request advisor help

Figure F31. Nearby mapped hotels, transit stops and attractions from approximate GPS

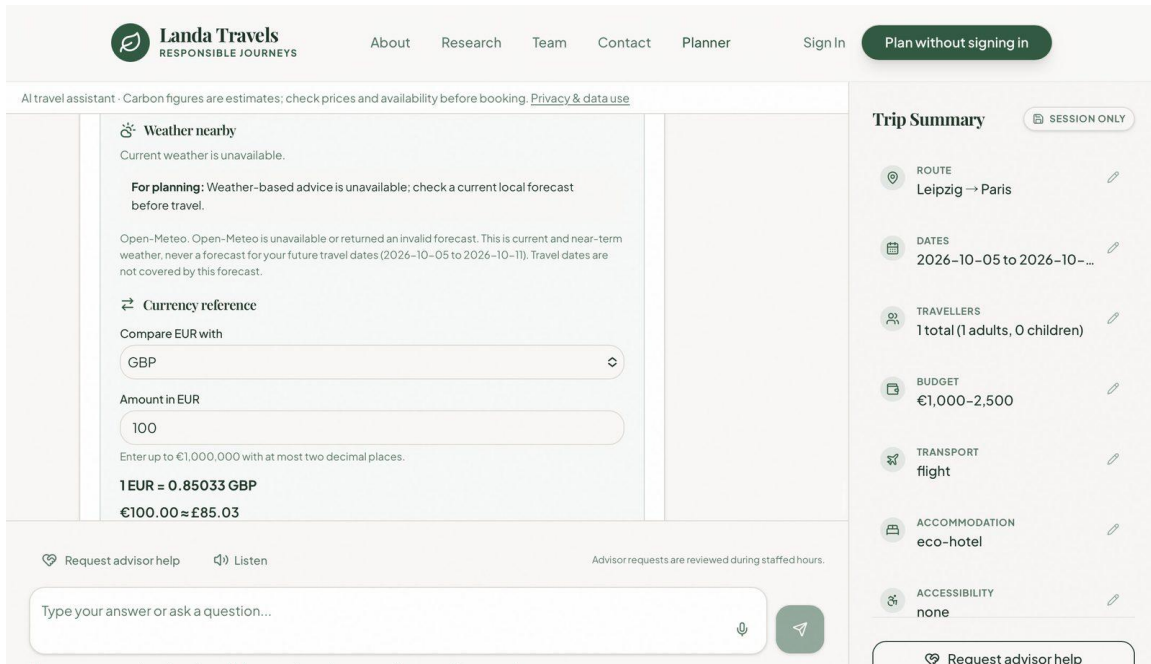


Figure F32. Weather and currency reference panels

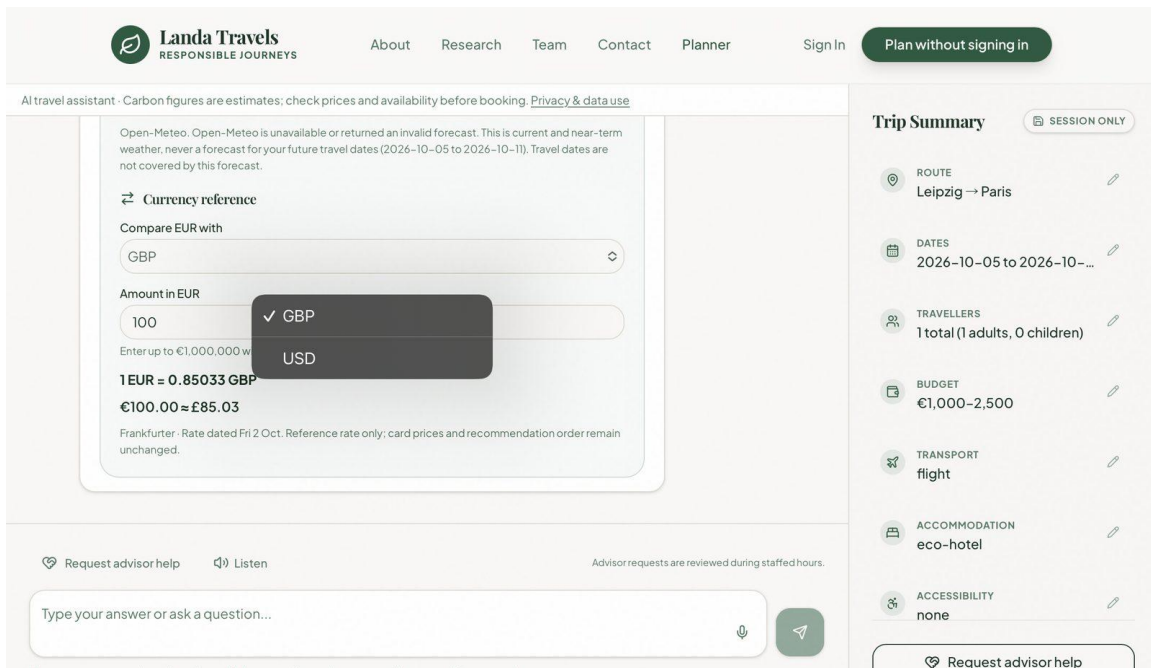


Figure F33. Currency reference selector

F.5 Human help



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Open-Meteo. Open-Meteo is unavailable or returned an invalid forecast. This is current and near-term weather, never a forecast for your future travel dates (2026-10-05 to 2026-10-11). Travel dates are not covered by this forecast.

↻ Currency reference

Compare EUR with

GBP

Amount in EUR

100

Enter up to €1,000,000 with at most two decimal places.

1 EUR = 0.85033 GBP

€100.00 ≈ £85.03

Frankfurt · Rate dated Fri 2 Oct. Reference rate only; card prices and recommendation order remain unchanged.

Request a later advisor reply

This is not live chat. An advisor reviews requests during staffed hours and replies later; timing depends on availability.

We will store a copy of your trip details and displayed options, including selected functional access requirements, and up to 30 recent messages after redacting contact details, precise locations, credentials, and sensitive free-text accessibility or medical information. This request context is visible only to its assigned advisor; the transcript is not included in the inbox notification email.

Email for your advisor's reply (no account required)

you@example.com

Your email is stored separately for this request and shared only with its assigned advisor and the email provider. You can also read the reply in this planning tab. Guest requests expire with their original conversation (up to 30 days).

☐ I agree to share and store the redacted conversation and privacy-minimised trip context with an assigned travel advisor, and to be contacted at the email above about

 Request advisor help
  Listen

Advisor requests are reviewed during staffed hours.




Figure F34. Human-help panel: not live chat, what is stored, reply email



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Open-Meteo. Open-Meteo is unavailable or returned an invalid forecast. This is current and near-term weather, never a forecast for your future travel dates (2026-10-05 to 2026-10-11). Travel dates are not covered by this forecast.

↻ Currency reference

Compare EUR with

GBP

Amount in EUR

100

Enter up to €1,000,000 with at most two decimal places.

1 EUR = 0.85033 GBP

€100.00 ≈ £85.03

Frankfurt · Rate dated Fri 2 Oct. Reference rate only; card prices and recommendation order remain unchanged.

We will store a copy of your trip details and displayed options, including selected functional access requirements, and up to 30 recent messages after redacting contact details, precise locations, credentials, and sensitive free-text accessibility or medical information. This request context is visible only to its assigned advisor; the transcript is not included in the inbox notification email.

Email for your advisor's reply (no account required)

you@example.com

Your email is stored separately for this request and shared only with its assigned advisor and the email provider. You can also read the reply in this planning tab. Guest requests expire with their original conversation (up to 30 days).

☐ I agree to share and store the redacted conversation and privacy-minimised trip context with an assigned travel advisor, and to be contacted at the email above about this request.

Send advisor request

Start the conversation and enter a valid email, name, WhatsApp number and email.

Request advisor help

Listen

Advisor requests are reviewed during staffed hours.

Type your answer or ask a question...

📍

➤

Figure F35. Consent checkbox and disabled send button

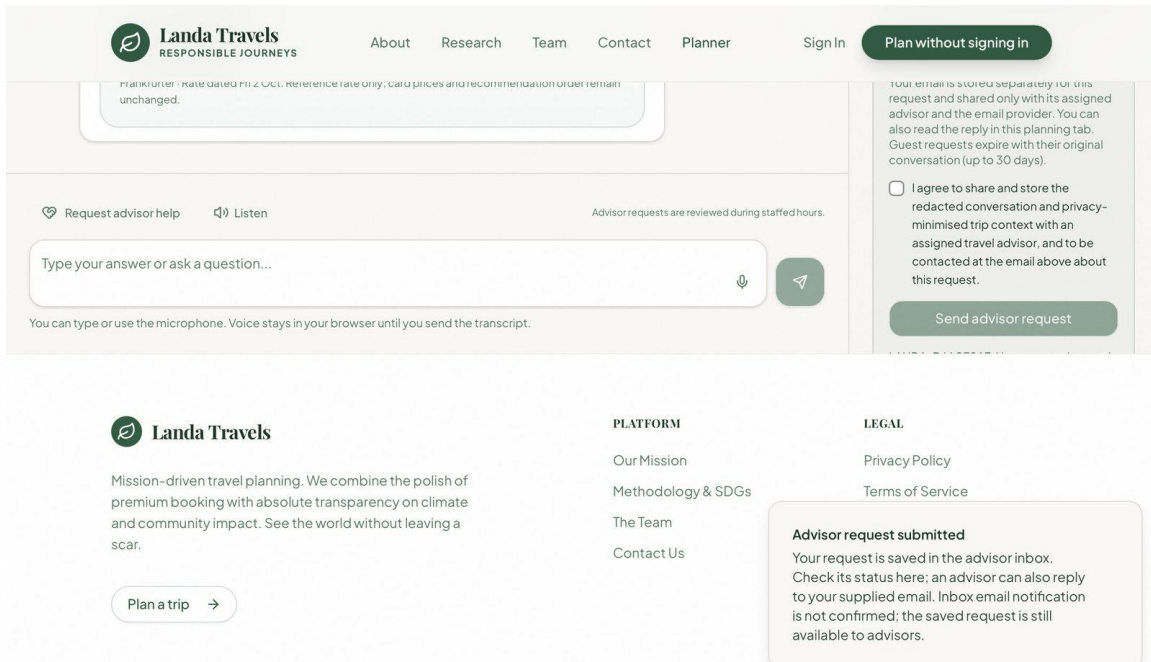


Figure F36. Confirmation after an advisor request is submitted

F.6 Account and rewards

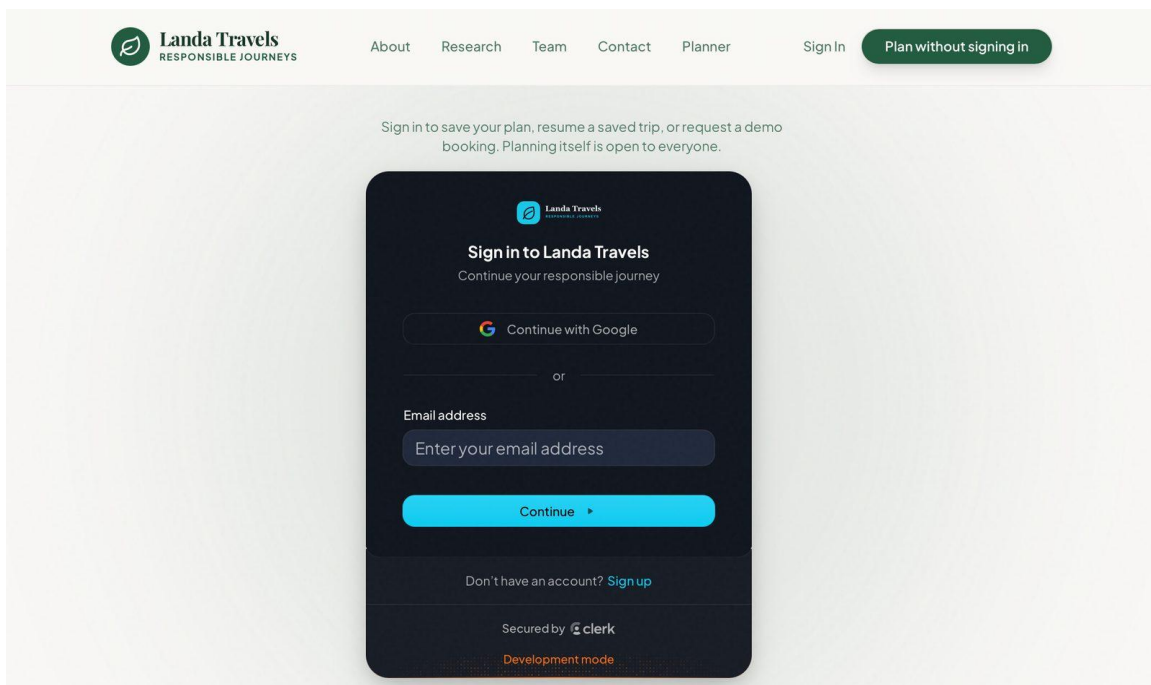


Figure F37. Sign-in page, with planning open to everyone

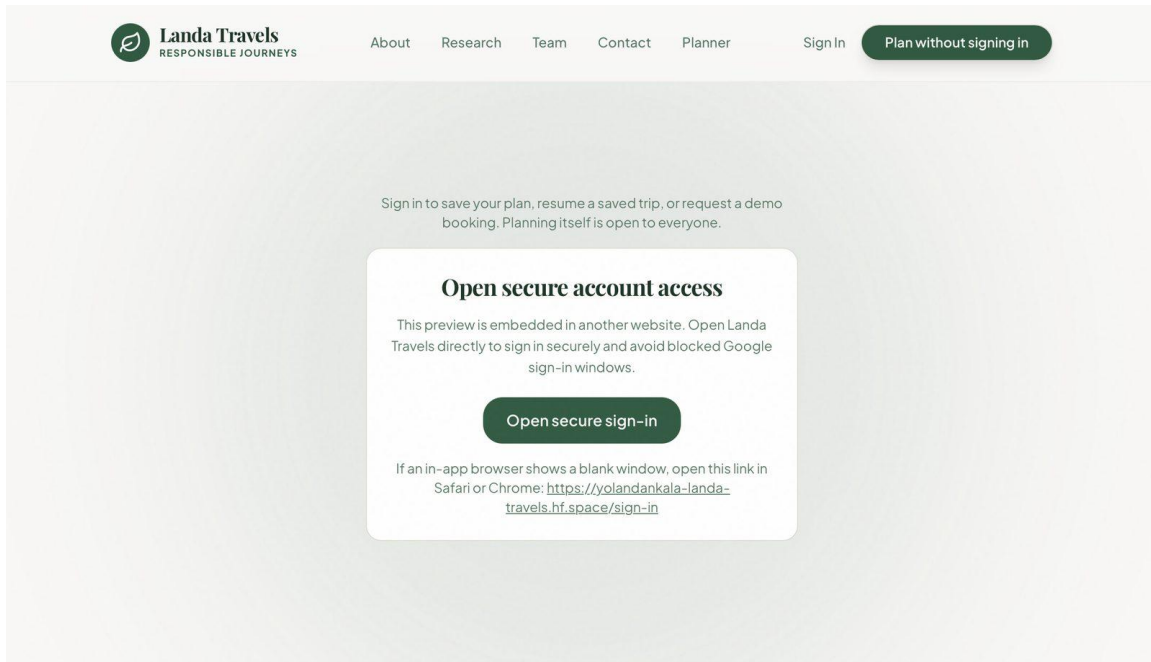


Figure F38. Secure sign-in prompt inside the Hugging Face preview

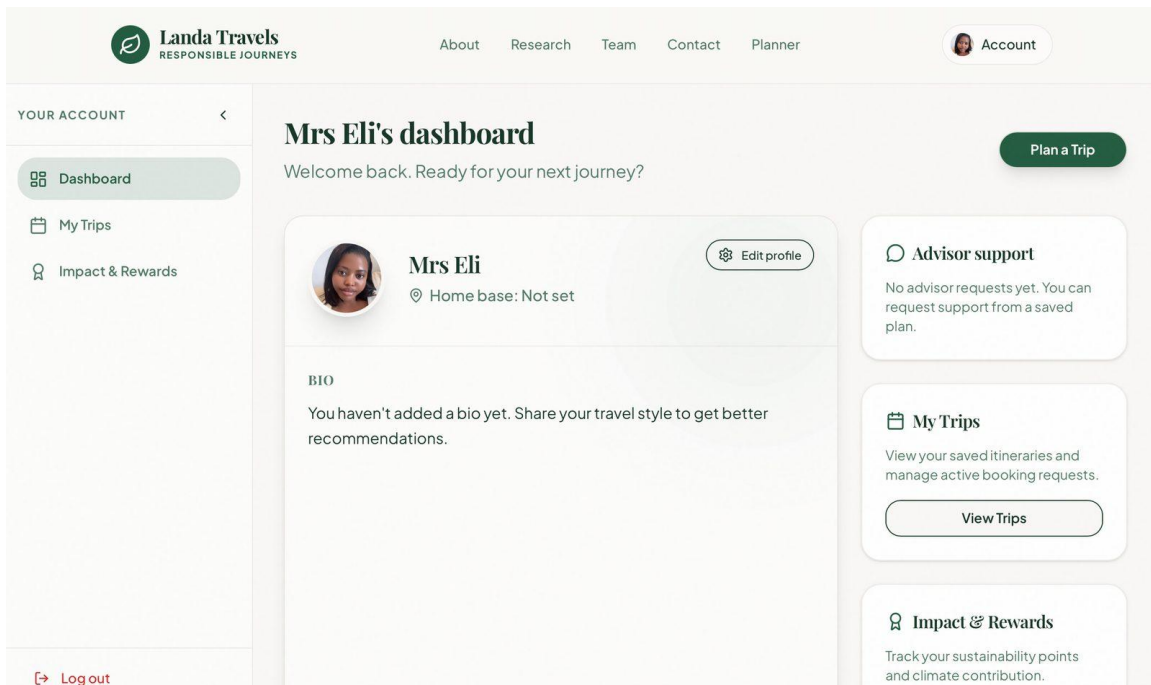


Figure F39. Traveller dashboard

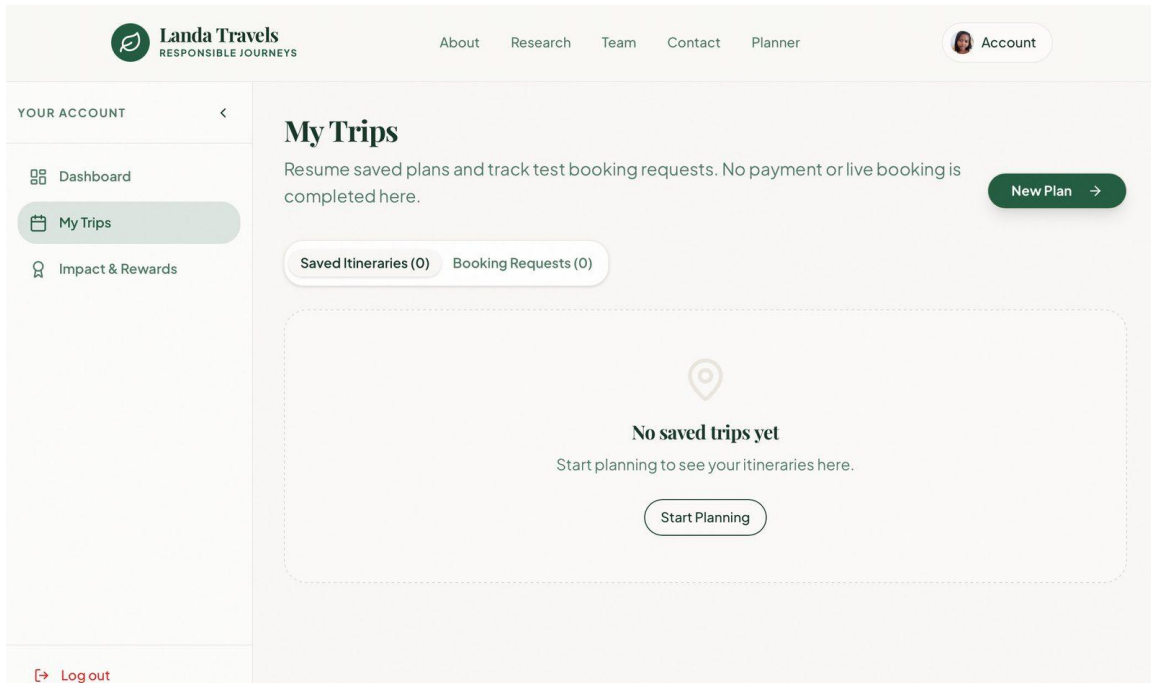


Figure F40. My Trips: saved itineraries and test booking requests

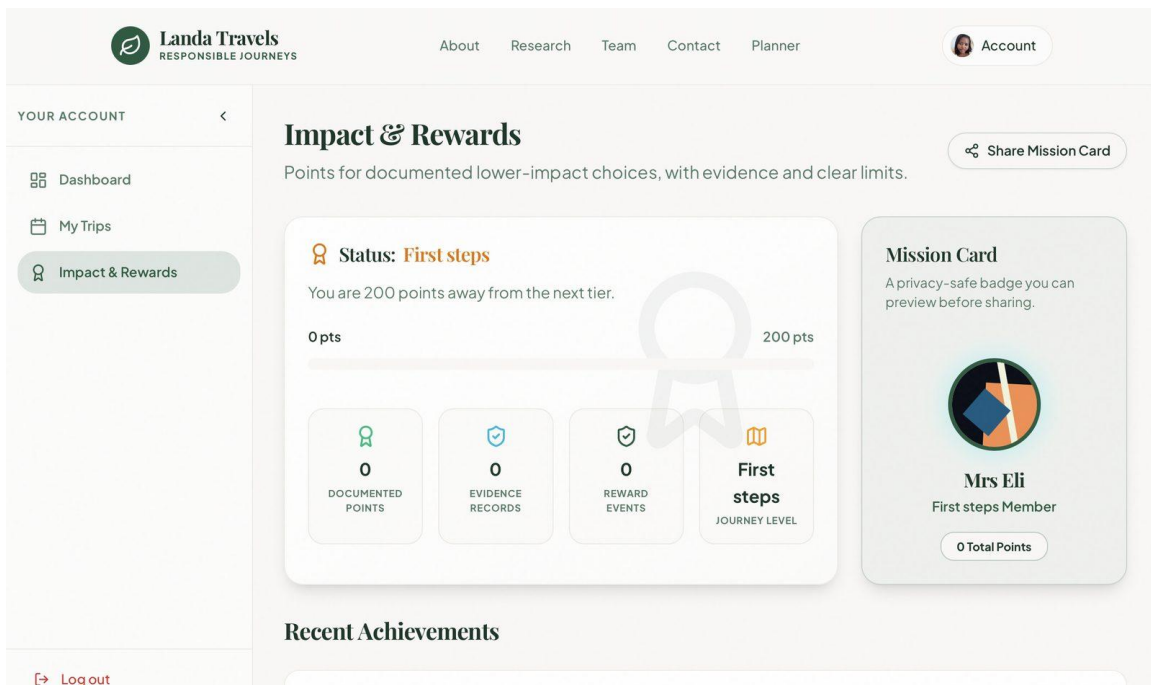


Figure F41. Impact & Rewards

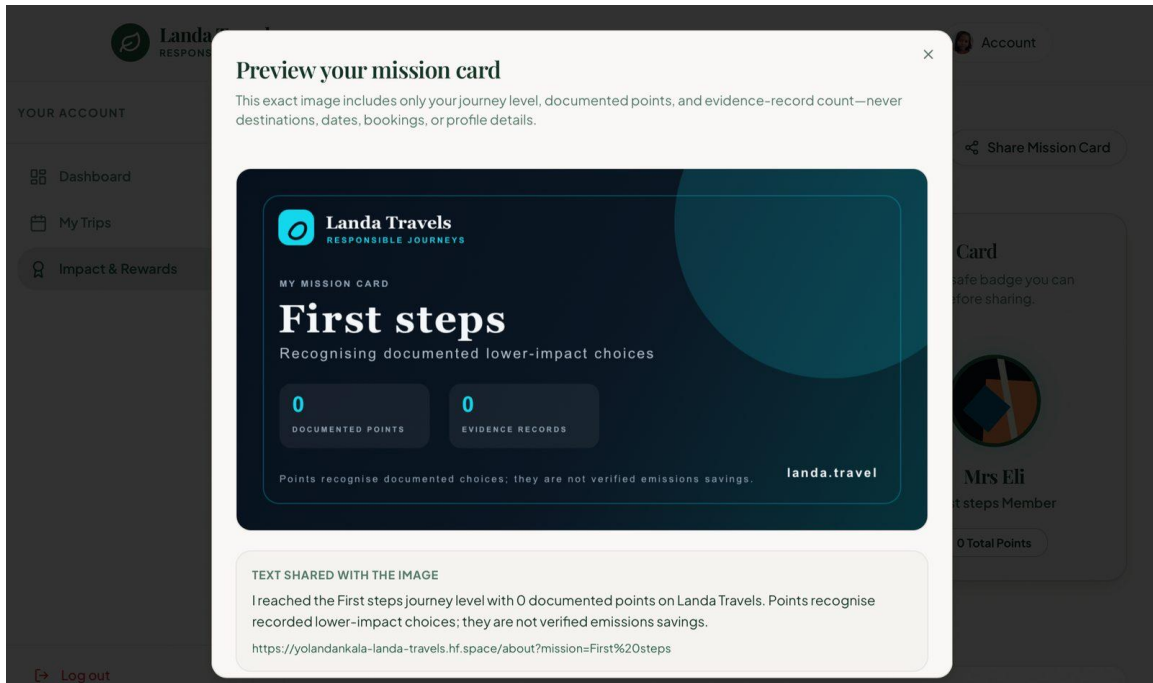


Figure F42. Privacy-safe mission card preview before sharing